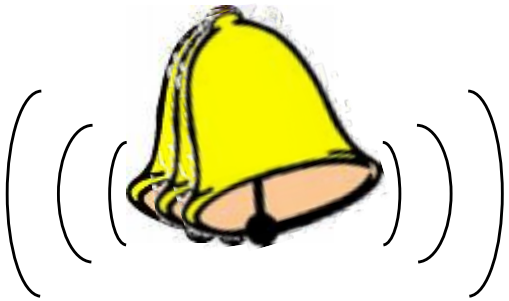
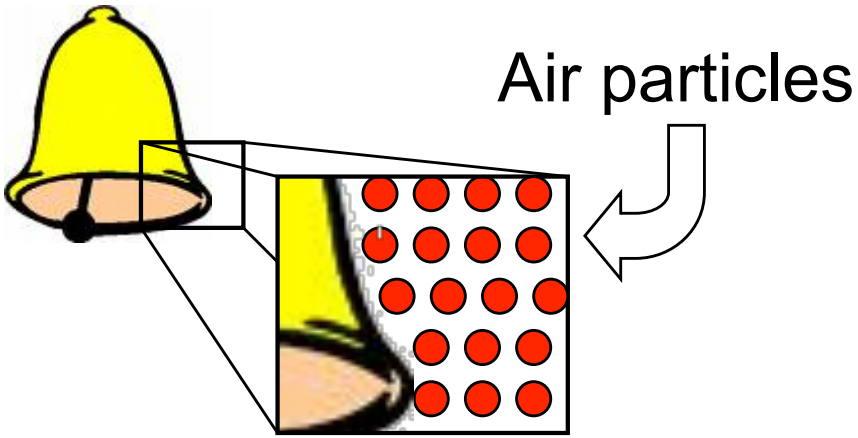
Fundamentals of Sound and Time-Frequency Representations

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Sound

- Sound is produced by a vibrating source that causes the matter around it to move.
- No sound is produced in a vacuum - Matter (air, water, earth) must be present

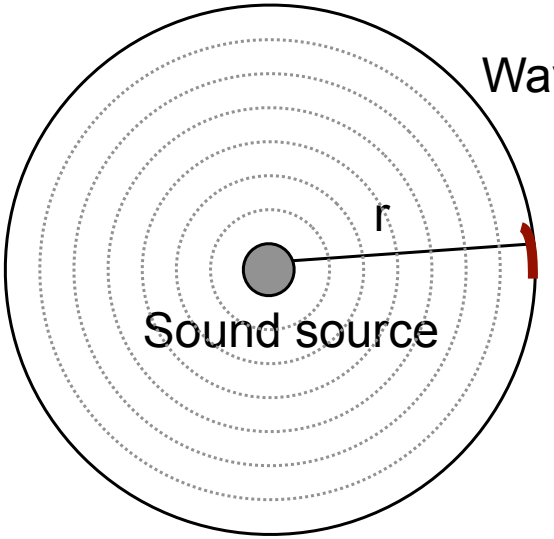




- The vibration of the source causes it to push/pull its neighboring particles, which in turn push/pull its neighbors and so on.
- Pushes increase the air pressure (compression) while pulls decrease the air pressure (rarefaction)
- The vibration sends a wave of pressure fluctuation through the air

Sound power and intensity

- A source (e.g. bell) vibrates when a force (e.g. striking hammer) is applied to it.
- The force applied and the resulting movement characterize the work performed by the source ($W = F \times \Delta s$)
- Power ($P = W/t$) is the rate at which work is performed and is measured in watts.
- An omnidirectional sound source produces a 3-D longitudinal wave. The resulting wavefront is defined by the surface of a sphere ($S = 4\pi r2$), where r is the distance from the source.



The original power is distributed on the surface of the wavefront.
As r increases, the power per unit area (intensity) decreases: $I = P/S$

Intensity and SPL

- The effect of sound power on its surroundings can be measured in sound pressure levels (SPL) - much as temperature in a room relates to the energy produced by a heater.
- Both intensity (Watts/area) and sound pressure (Newtons/area) are usually represented using decibels (dB)
- dB are based on the logarithm of the ratio between two powers, thus describing how they compare ($dB = 10\log_{10}(P1/P2)$).
- This can be applied to other measures (amplitude, SPL, voltage), as long as their relationship to power is taken into account.
- In the case of intensity and SPL, the denominator of the ratio is a reference value, defined according to the quietest sound perceivable by the average person.
- Thus by convention, 0 dB corresponds to SPL = $2x10^{-5}$ N/m^2 or I = 10^{-12} $watt/m^2$

Sound waves (1)

- In sound wave motion air particles do not travel, they oscillate around a point in space.

- The rate of this oscillation is known as the frequency (f) of the sound wave and is denoted in cycles per second (cps) or hertz (Hz).

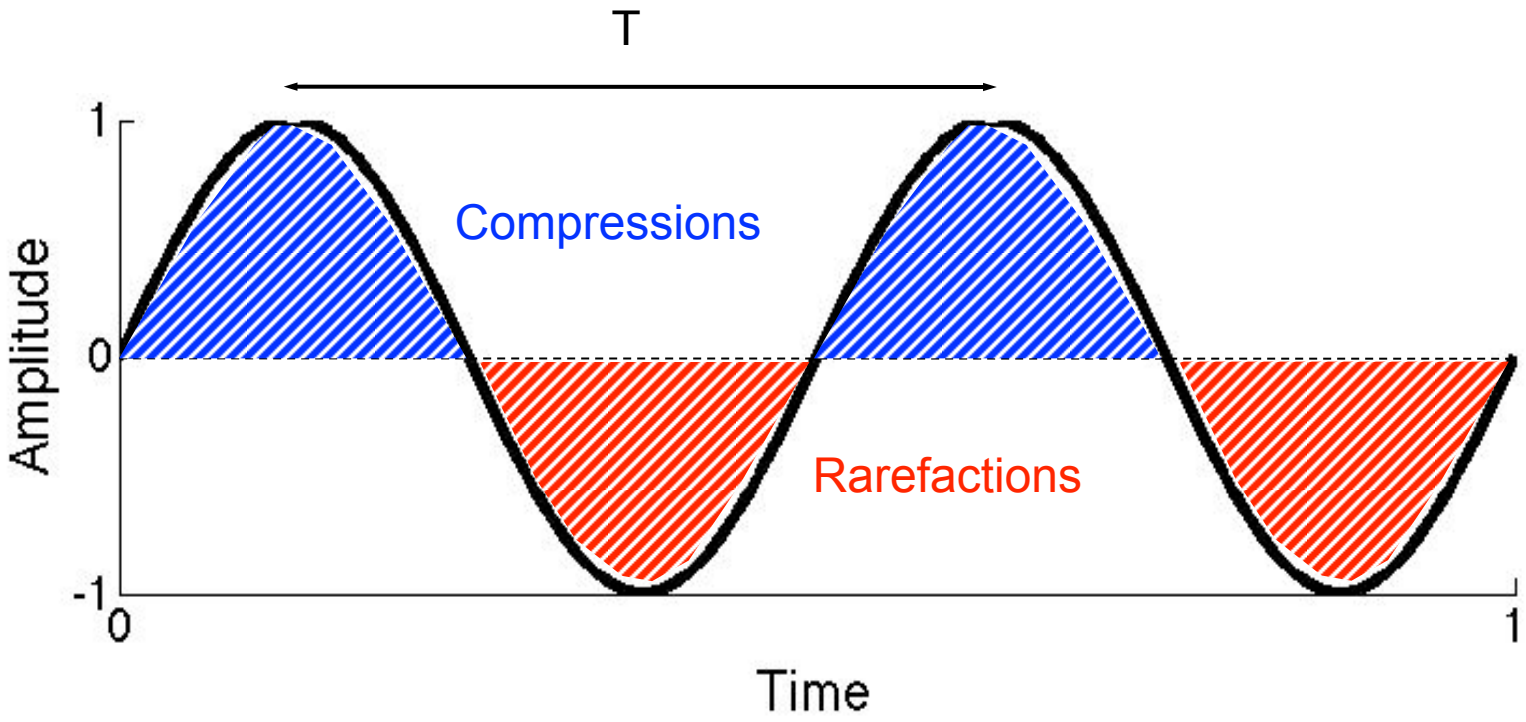
- The amount of compression/rarefaction of the air is the amplitude (A) of the sound wave.

- The distance between consecutive peaks of compression or rarefaction is the wavelength of the sound wave (λ)

Sound waves (2)

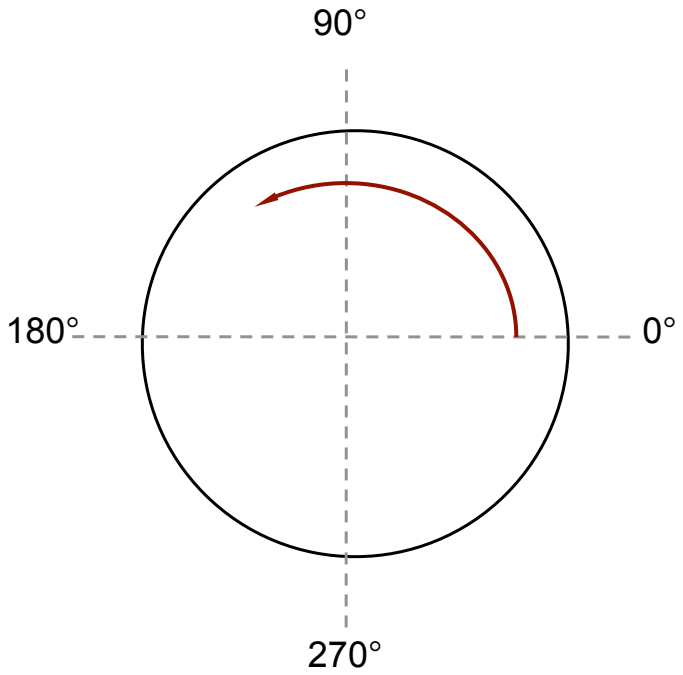
- If the frequency of the oscillation is stable, then the sound wave is periodic (with period T, and frequency f = 1/T)

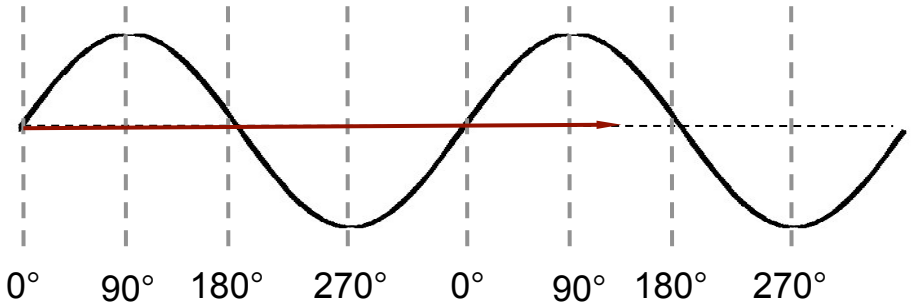
- The simplest periodic wave is a sinusoid: $x(t) = A \cdot sin(2\pi ft + \theta)$



Sound waves (3)

- Phase is a temporal offset, defined in terms of a fraction (degrees) of a complete cycle of the periodic wave.

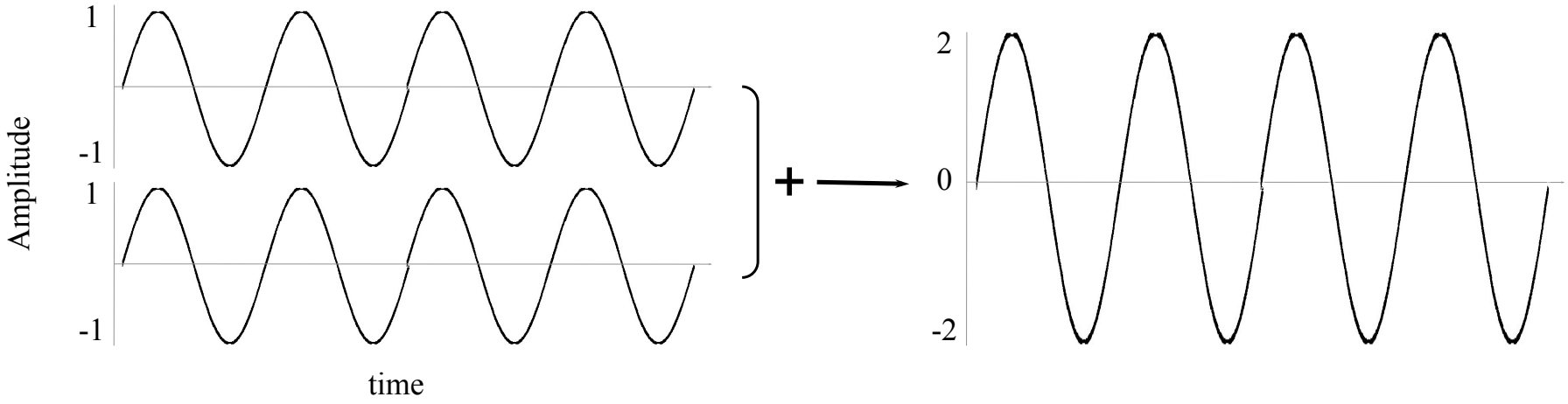




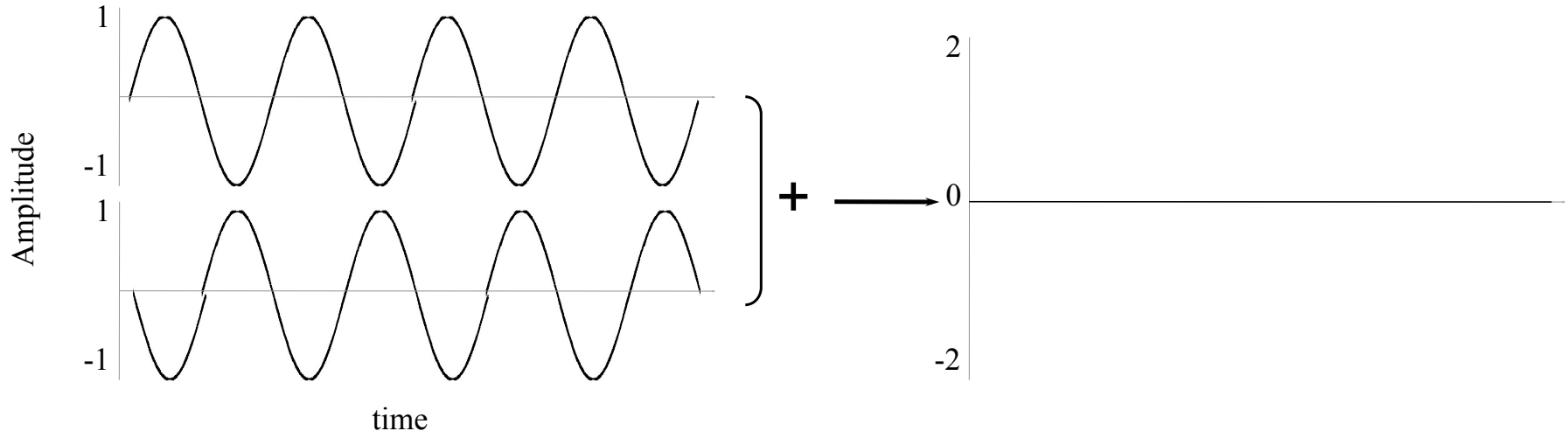
- The frequency defines the number of cycles per second, thus the time x frequency x 360° returns the (unwrapped) angular phase

Phase (1)

- In phase: cycles coincide exactly (sum duplicates amplitude)

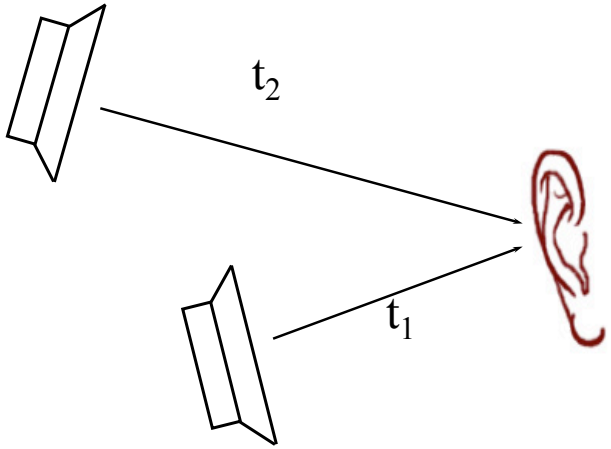


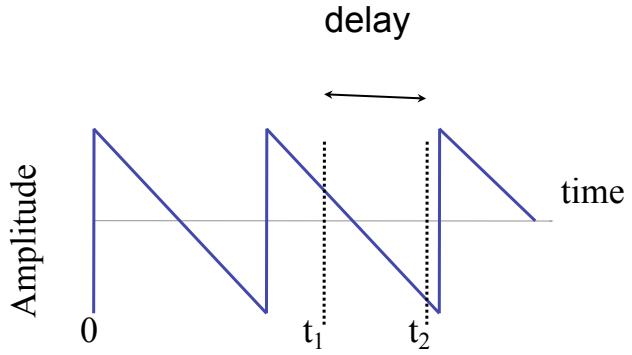
- Out of phase: half cycles are exactly opposed (sum cancels them)



Phase (2)

- There is a range of partial additions and cancellations in between those extremes
- What causes phase difference?

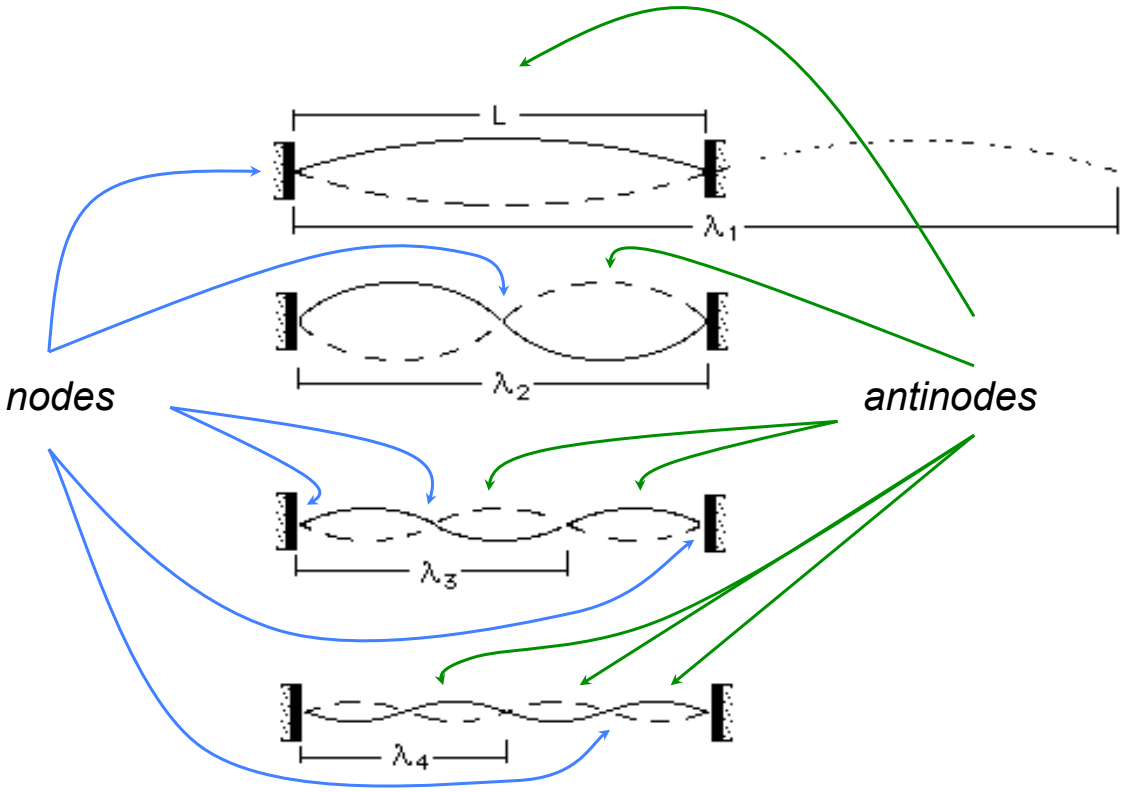




- The phase difference depends on the time deviation and the wave's frequency

Types of sounds (1)

- Sinusoids are only one possible type of sound corresponding to the simplest mode of vibration, producing energy at only one frequency
- Most sources are capable of vibrating in several harmonic modes at the same time, generating energy at different frequencies



1st harmonic

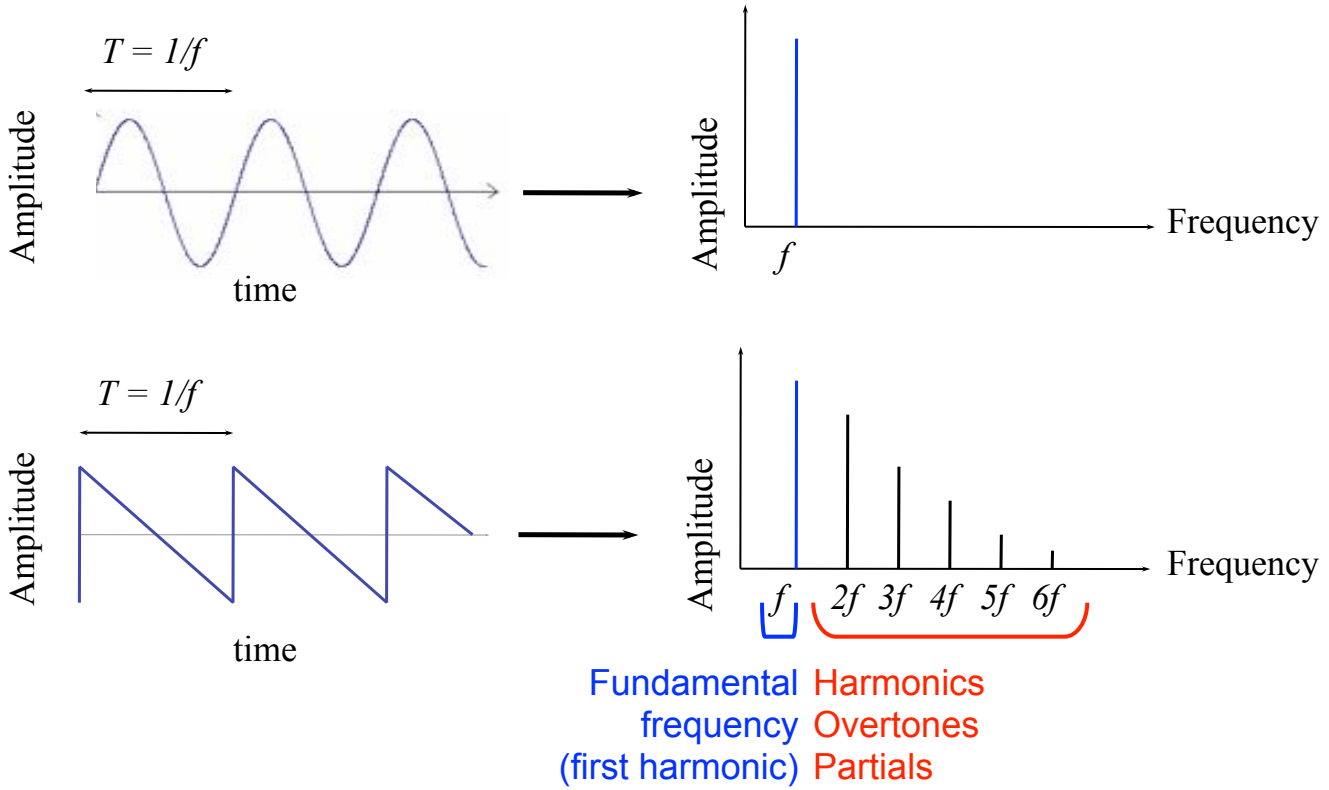
2nd harmonic

3rd harmonic

4th harmonic

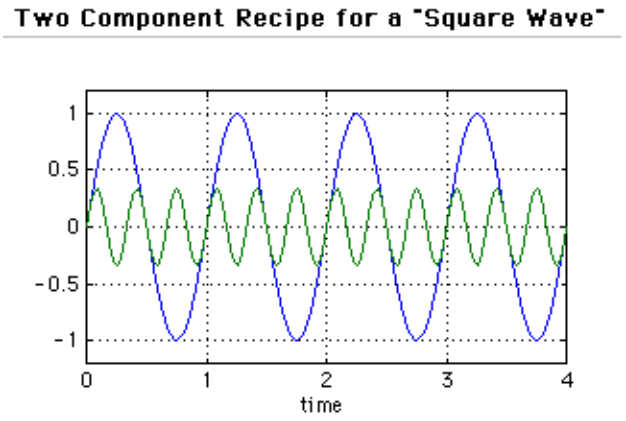
Types of sounds (2)

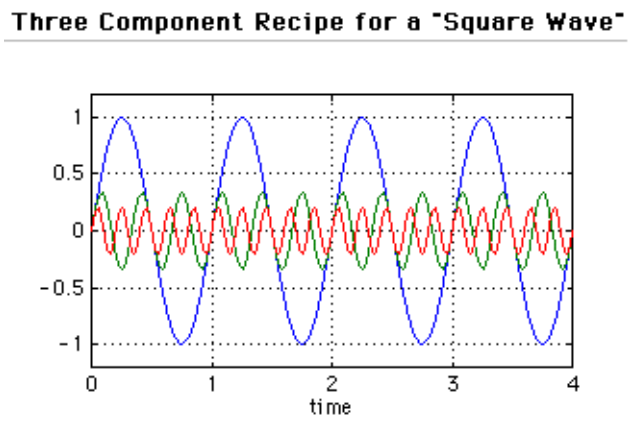
- Harmonics (or Overtones or Partials) are frequency components that occur at integer multiples of the fundamental frequency
- Their amplitude variations determine the timbre of the sound

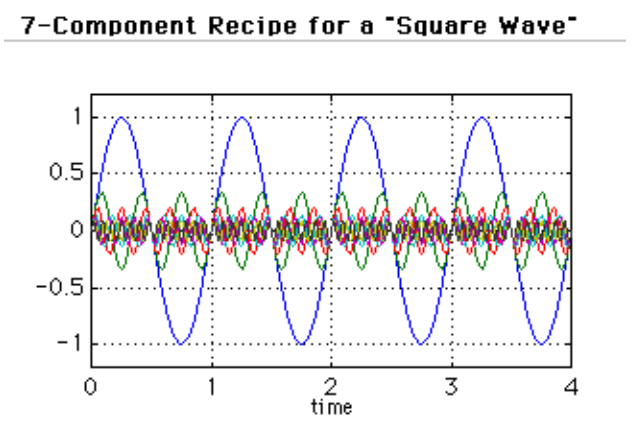


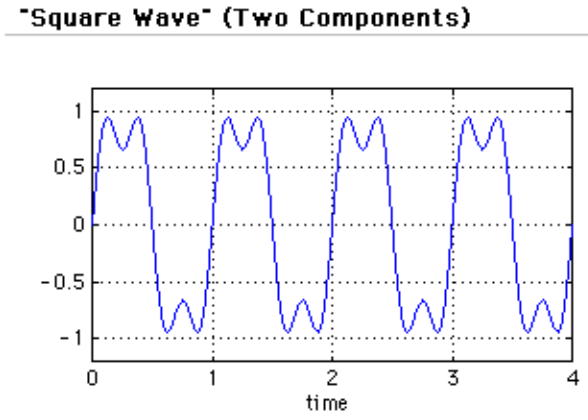
Types of sounds (3)

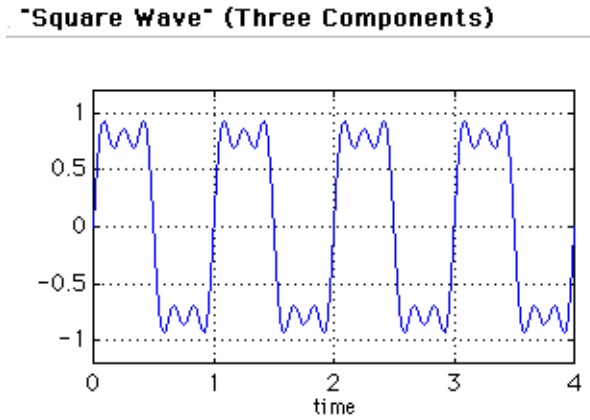
- Example: Square wave - only odd harmonics (even are missing). Amplitude of the nth harmonic = 1/n

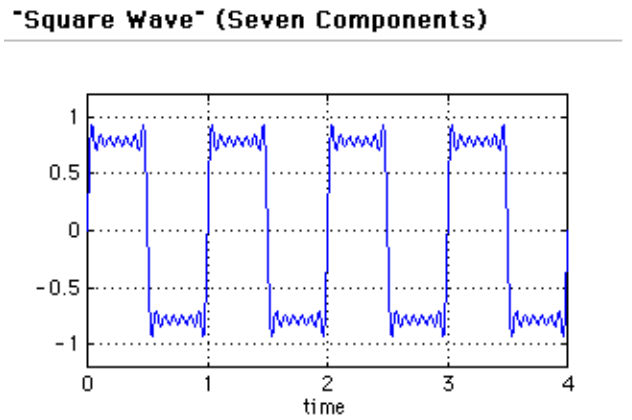






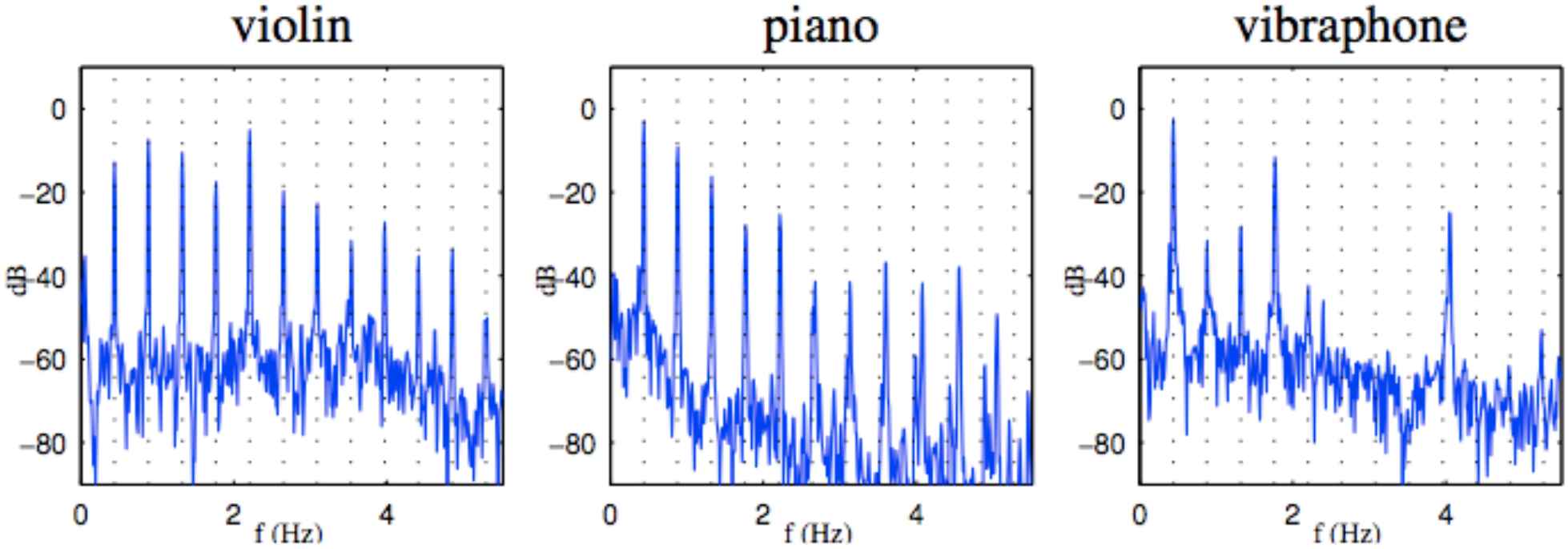






Types of sounds (4)

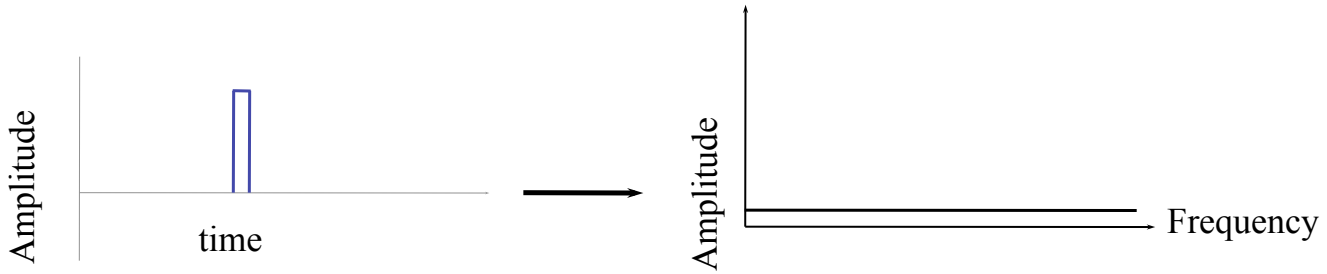
- Most natural pitched sounds also present overtones which are not integer multiples of the fundamental.
- These are known as inharmonic partials



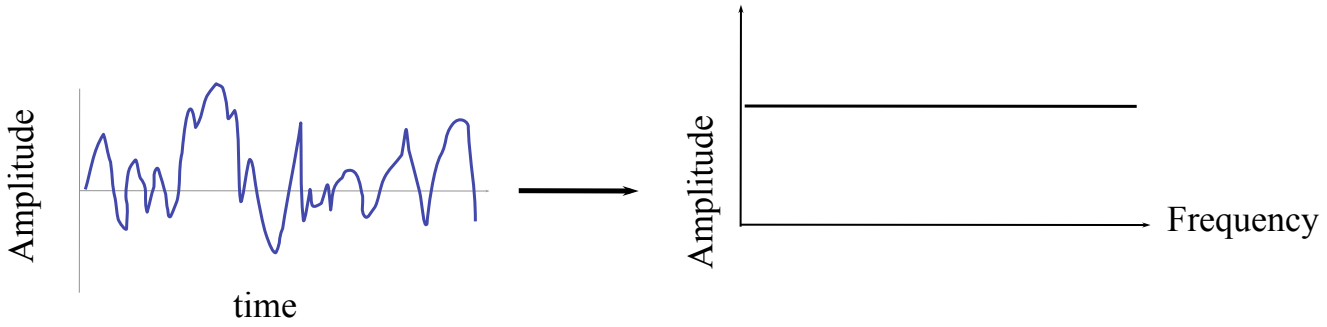
Harmonic ⟵ ⟶ Inharmonic

Types of sounds (5)

- Non-periodic sounds have no pitch and tend to have continuous spectra, e.g. a short pulse (narrow in time, wide in frequency)

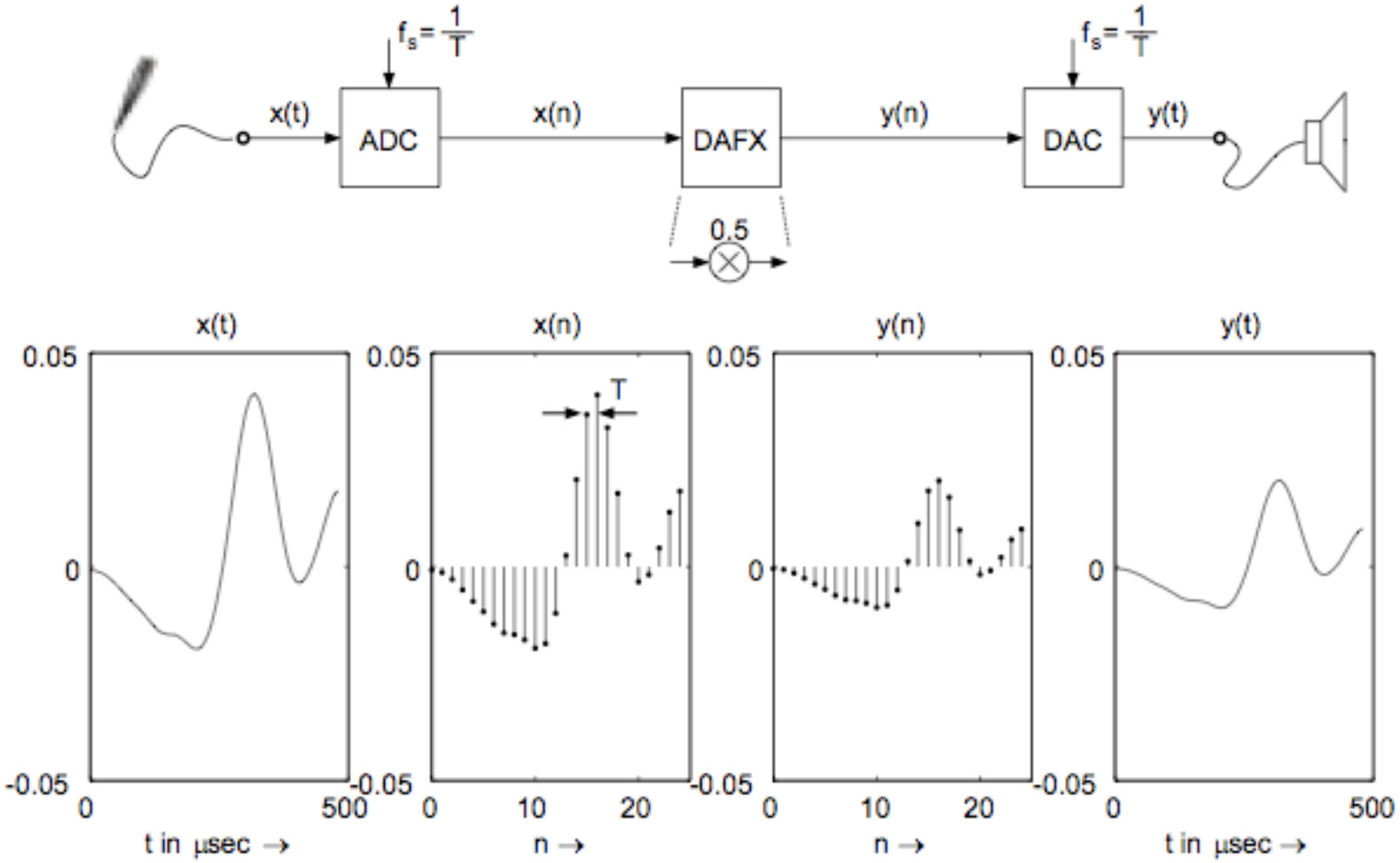


- The most complex sound is white noise (completely random)

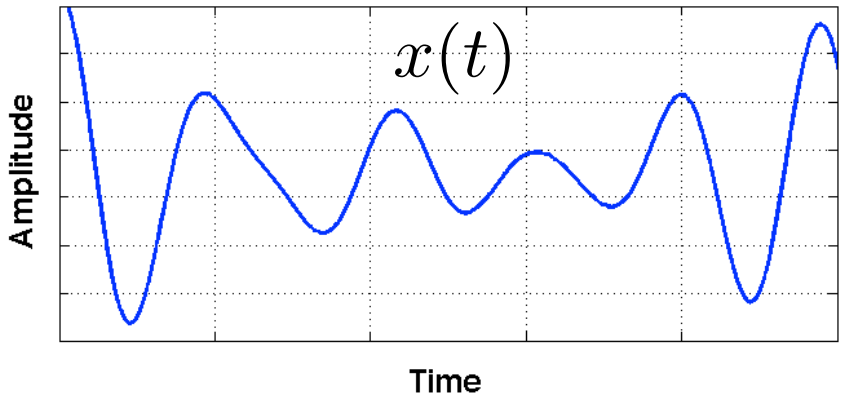


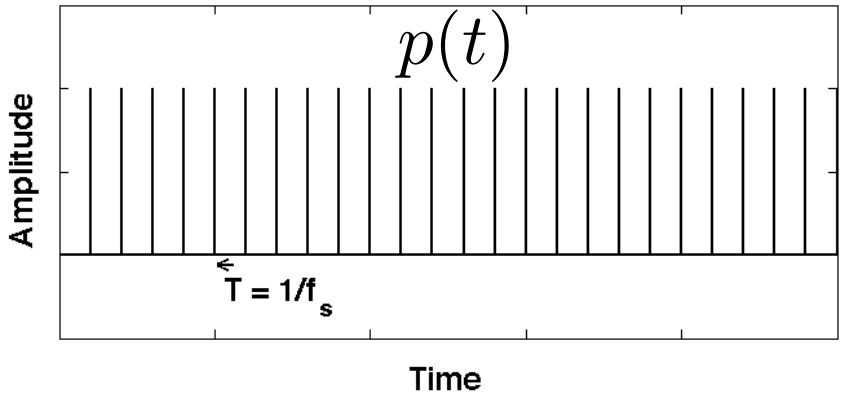
- The more complex, the noisier the sound is

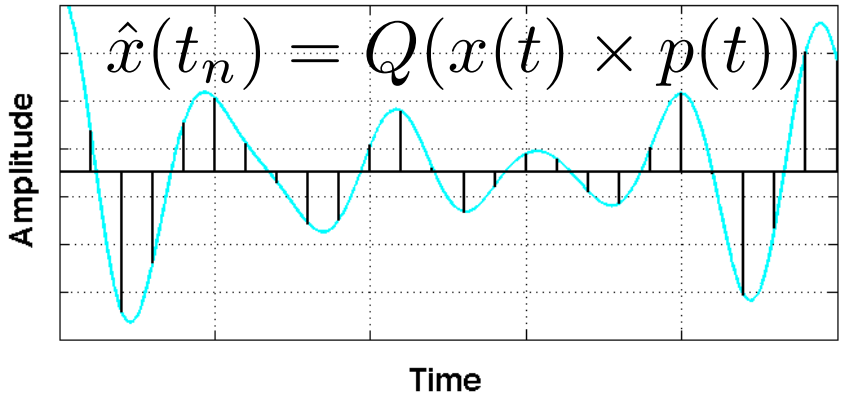
Digital audio

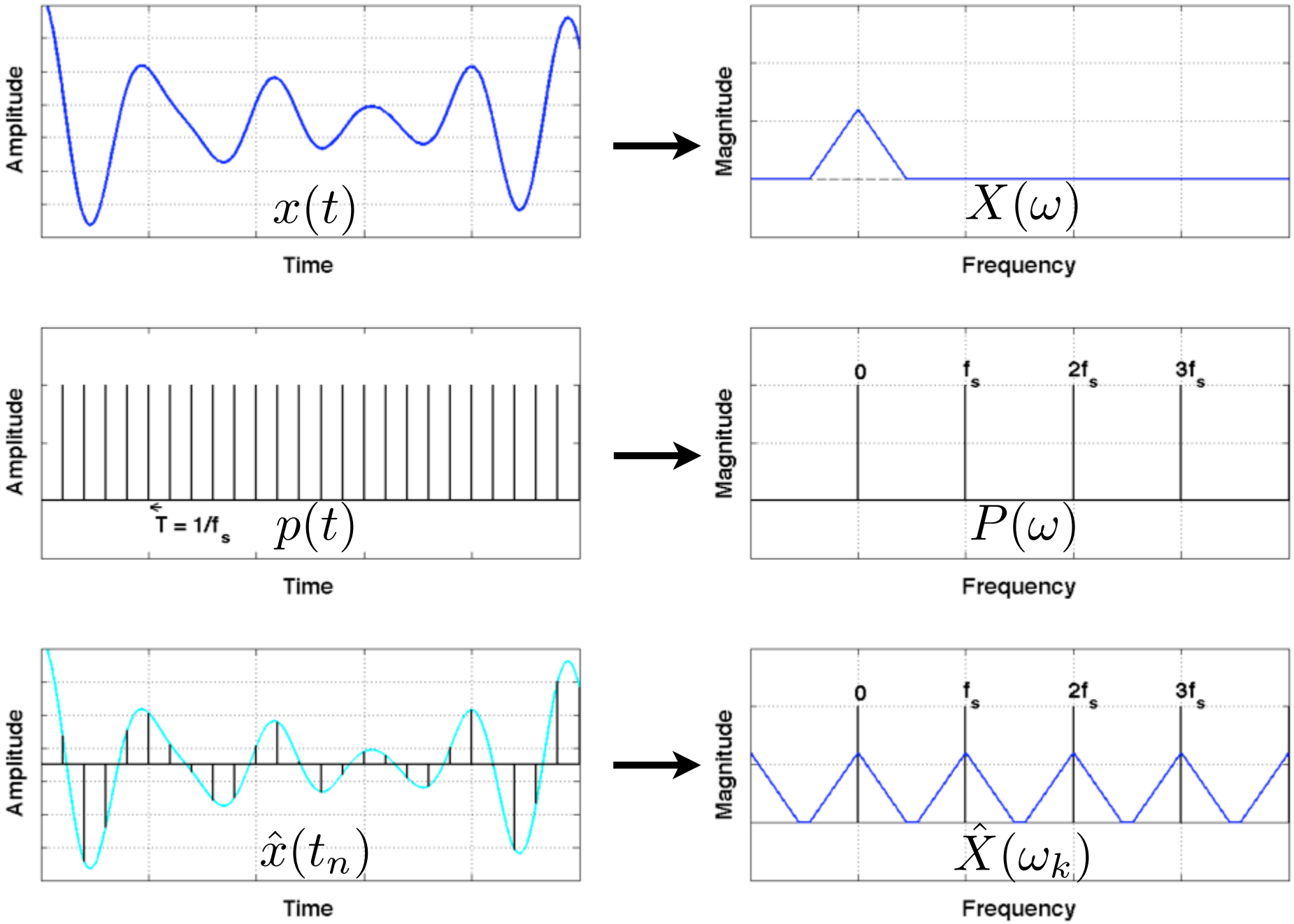


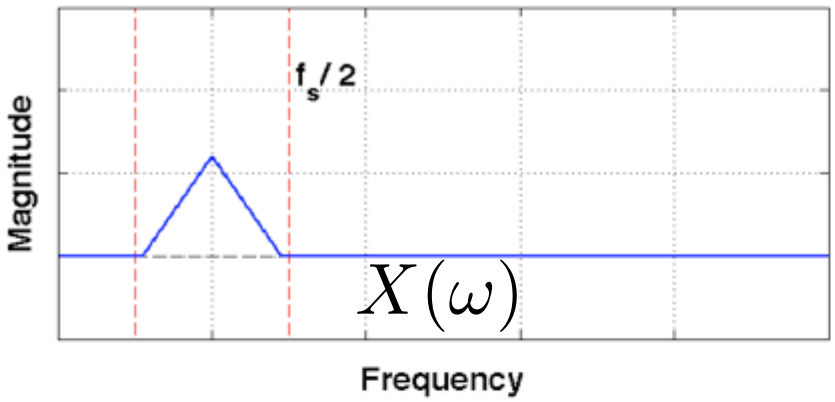
Discrete Signal and Sampling

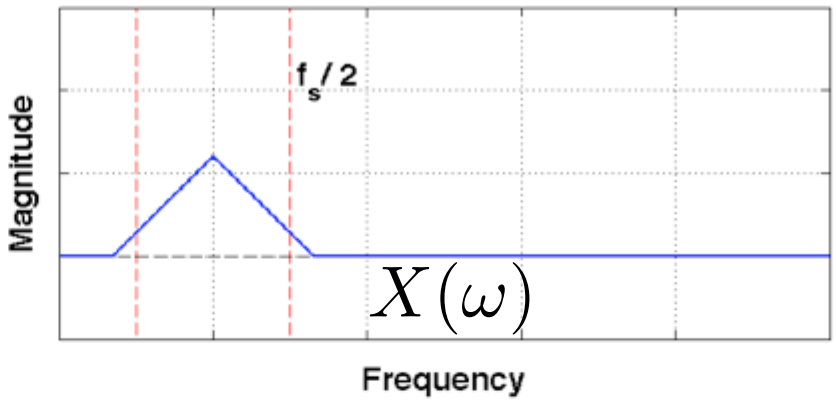


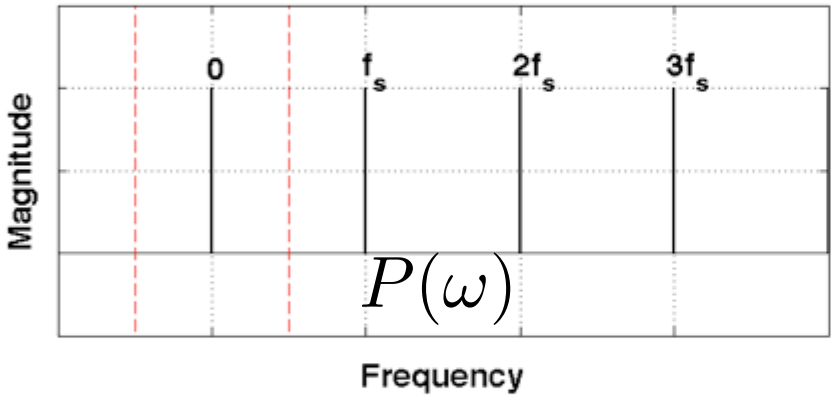


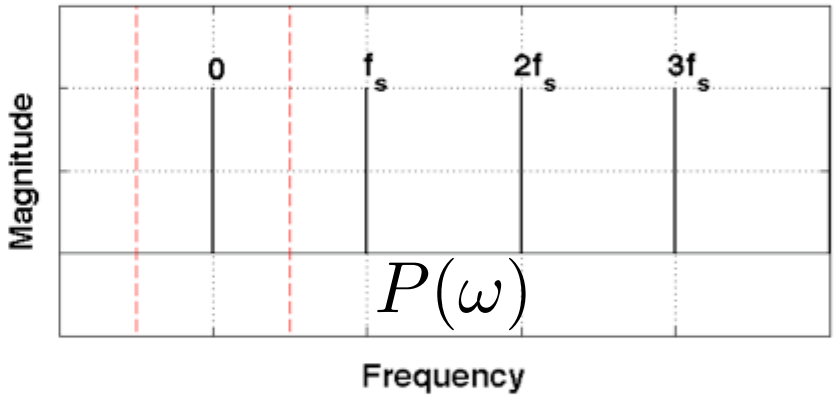


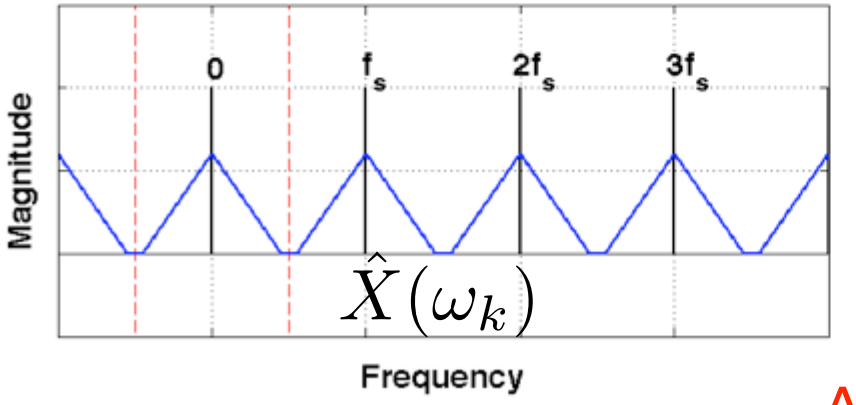

Amplitude
$x(t)$
Time
Magnitude
$X(\omega)$
Frequency
Amplitude
T = 1/f_s
$p(t)$
Time
Magnitude
0
f_s
2f_s
3f_s
$P(\omega)$
Frequency
Amplitude
$\hat{x}(t_n)$
Time
Magnitude
0
f_s
2f_s
3f_s
$\hat{X}(\omega_k)$
Frequency

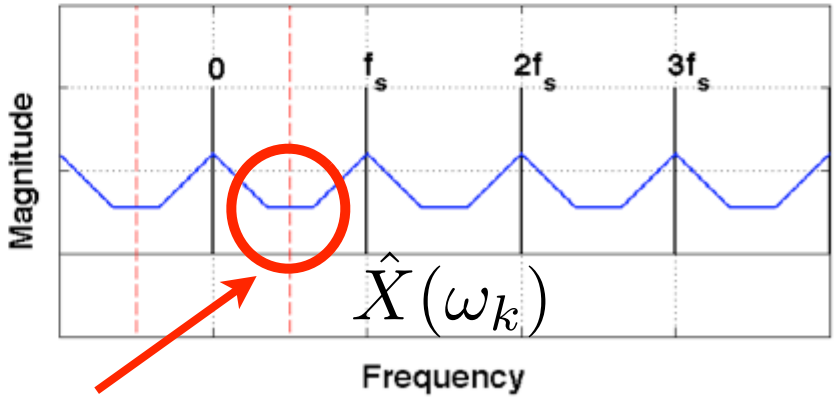

Magnitude
$f_s/2$
$X(\omega)$
Frequency


Magnitude
$f_s/2$
$X(\omega)$
Frequency


Magnitude
0
f_s
$2f_s$
$3f_s$
$P(\omega)$
Frequency

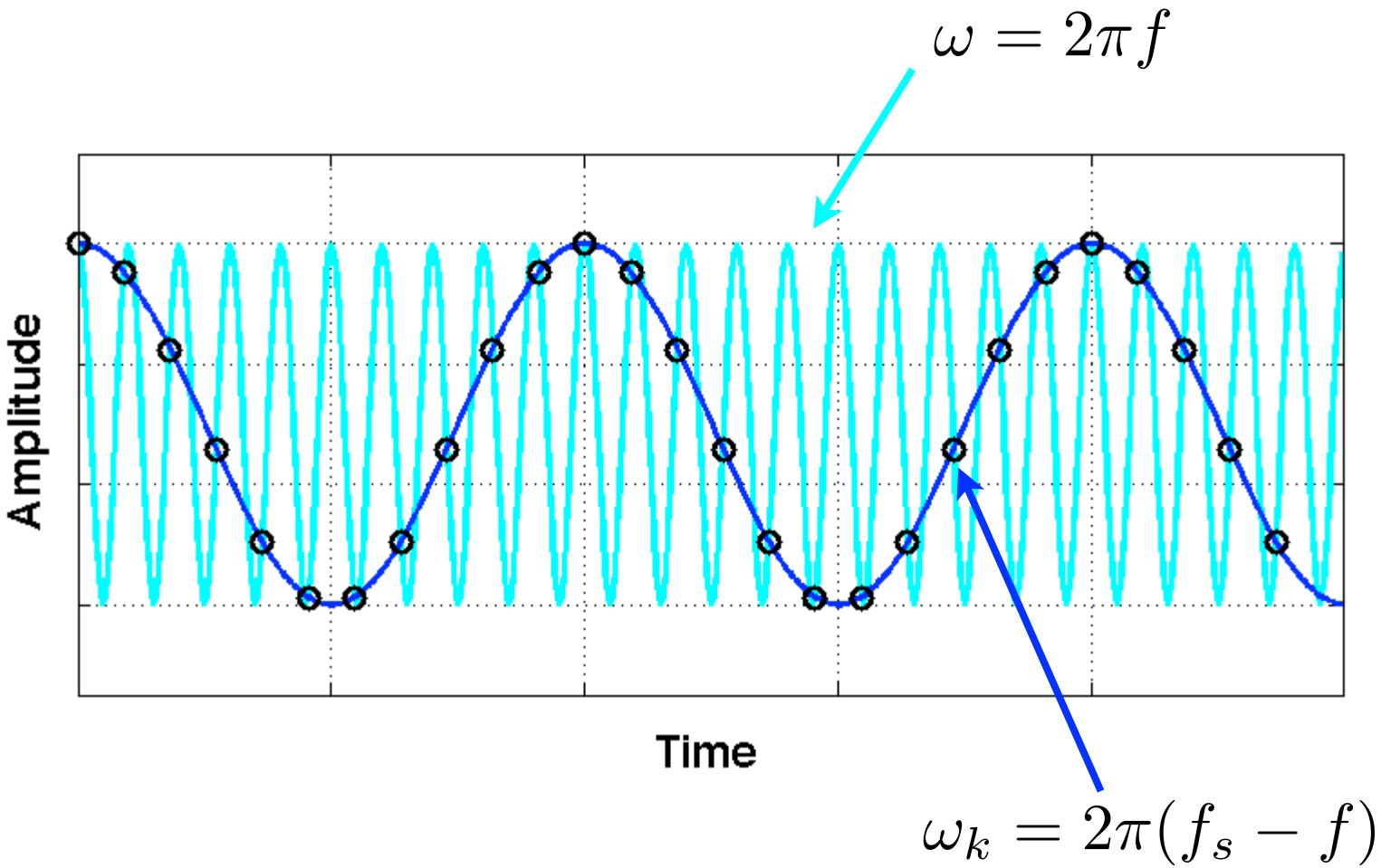

Magnitude
0
f_s
$2f_s$
$3f_s$
$P(\omega)$
Frequency


Magnitude
0
f_s
$2f_s$
$3f_s$
$\hat{X}(\omega_k)$
Frequency


Magnitude
0
f_s
$2f_s$
$3f_s$
$\hat{X}(\omega_k)$
Frequency

Aliasing

Aliasing



Discrete Fourier Transform (DFT)

$$X(\omega_k) \equiv \sum_{n=0}^{N-1} x(t_n) e^{-j\omega_k t_n}$$

x = input signal
$t_n = \frac{n}{fs}$ = discrete time (s), $n \geq 0$ is an integer
fs = sampling rate (Hz)
X = spectrum of x
$\omega_k = k\Omega$ = discrete frequency (rad/s), $k \geq 0$ is an integer
$\Omega = 2\pi(\frac{fs}{N})$ = frequency sampling interval (rad/s)
N = number of time/frequency samples

Simple form :

$$X(k) \equiv \sum_{n=0}^{N-1} x(n) e^{-j2\pi nk/N}$$

Discrete Fourier Transform (DFT)

This can also be written as:

$$X(k) = \langle x(n), s_k(n) \rangle$$

Which can be formulated as a matrix multiplication:

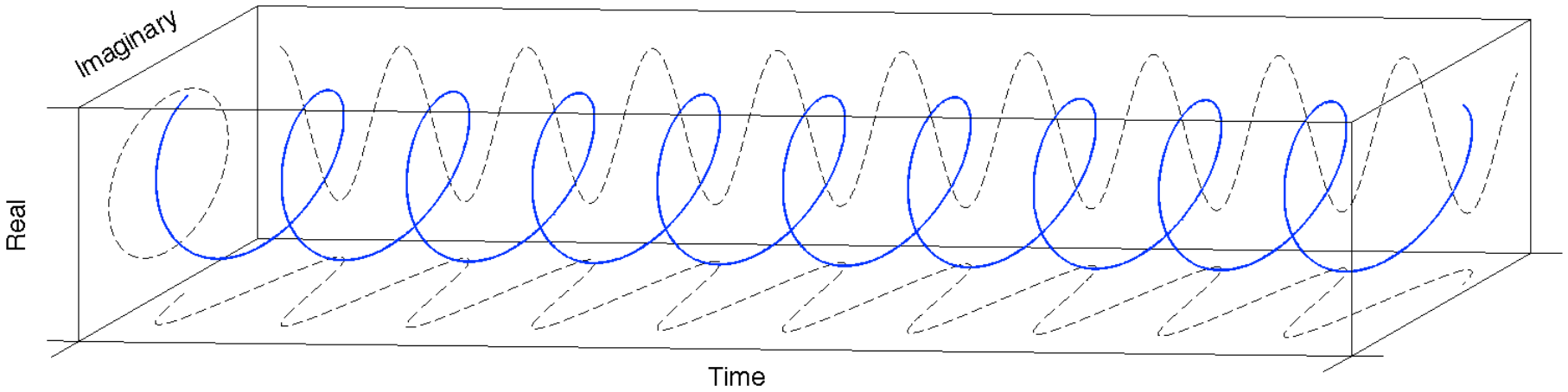
$$\begin{bmatrix} X(0) \\ X(1) \\ \vdots \\ X(N-1) \end{bmatrix} = \begin{bmatrix} s_0^*(0) & s_0^*(1) & \cdots & s_0^*(N-1) \\ s_1^*(0) & s_1^*(1) & \cdots & s_1^*(N-1) \\ \vdots & \vdots & \ddots & \vdots \\ s_{N-1}^*(0) & s_{N-1}^*(1) & \cdots & s_{N-1}^*(N-1) \end{bmatrix} \begin{bmatrix} x(0) \\ x(1) \\ \vdots \\ x(N-1) \end{bmatrix}$$

Discrete Fourier Transform (DFT)

where,

$$s_k(n) = e^{j2\pi nk/N} = \overbrace{cos(2\pi nk/N)}^{\text{Real part}} + \overbrace{jsin(2\pi nk/N)}^{\text{Imaginary part}}$$

is the set of the sampled complex sinusoids with a whole number of periods in N samples (Smith, 2007).



Discrete Fourier Transform (DFT)

The N resulting $X(k)$ are complex-valued vectors $X_R(k) + jX_I(k)$ such that, $\forall k = 0, 1, \cdots, N-1$:

$$|X(k)| = \sqrt{X_R^2(k) + X_I^2(k)}$$

$$\angle X = \phi(k) = tan^{-1} \frac{X_I(k)}{X_R(k)}$$

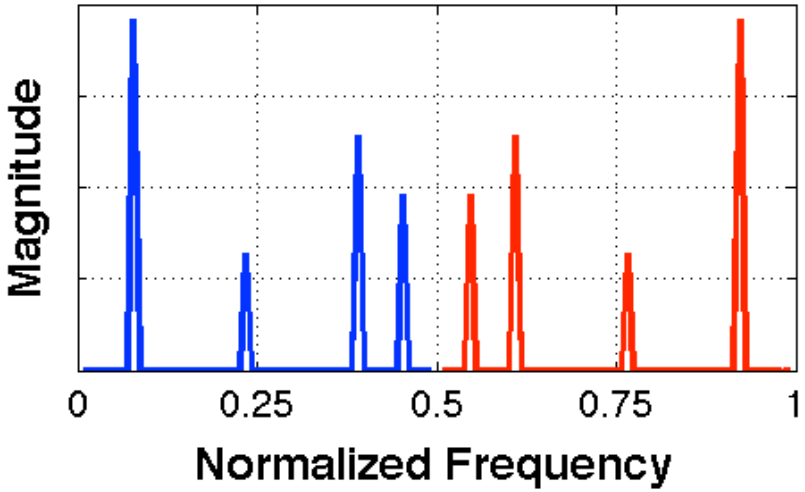
Furthermore, if $x(n)$ is real-valued, then:

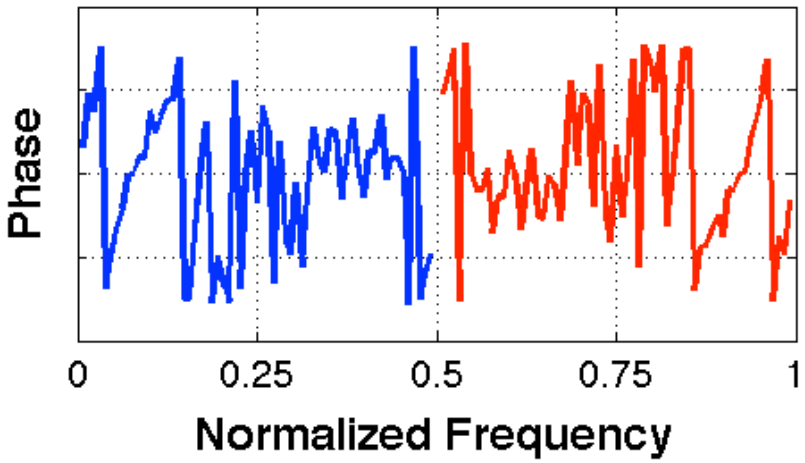
$$X(k) = X^*(N-k)$$

The DFT of an audio signal is half-redundant!

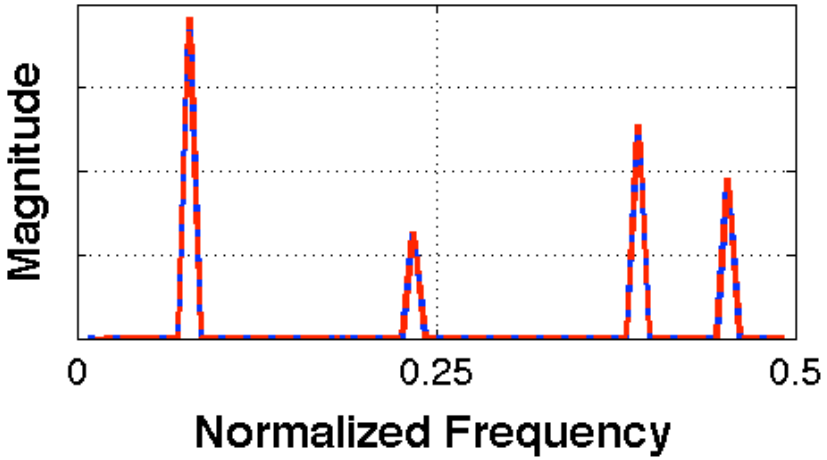
Discrete Fourier Transform (DFT)

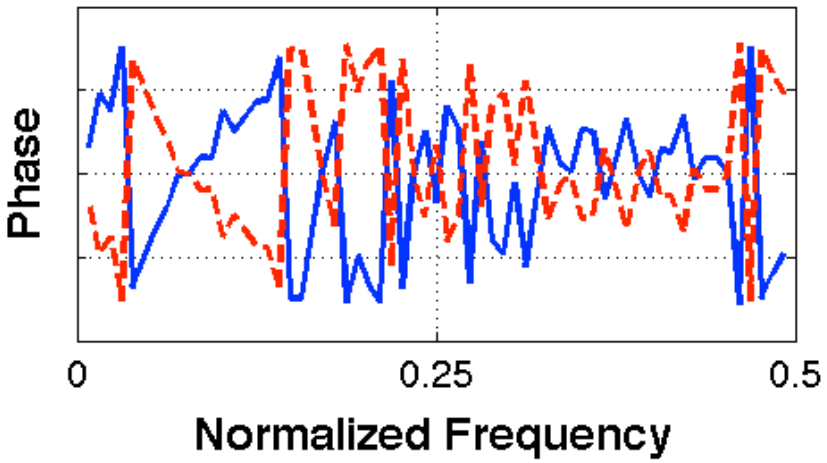
Spectrum





Conjugate Pairs





The IDFT and FFT

- The Inverse DFT (IDFT) is defined as:

$$x(n) \equiv \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{j2\pi nk/N}, \quad n = 0, 1, \cdots, N-1$$

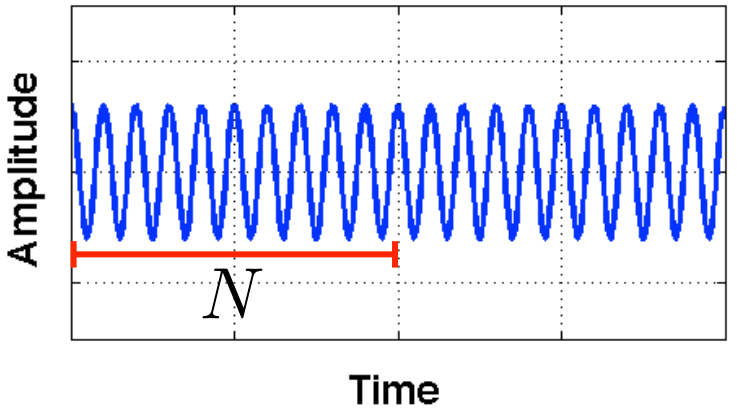
- The DFT needs on the order of N^2 operations for its computation.

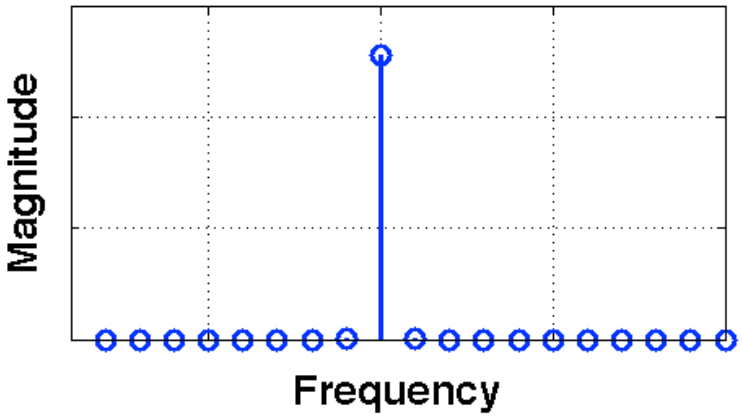
- The Fast Fourier Transform (FFT) is an efficient implementation of the DFT, that only requires on the order of $N\log_2 N$ operations when N is a power of 2.

- The FFT is so fast that it can be used to efficiently perform time-domain operations such as convolution.

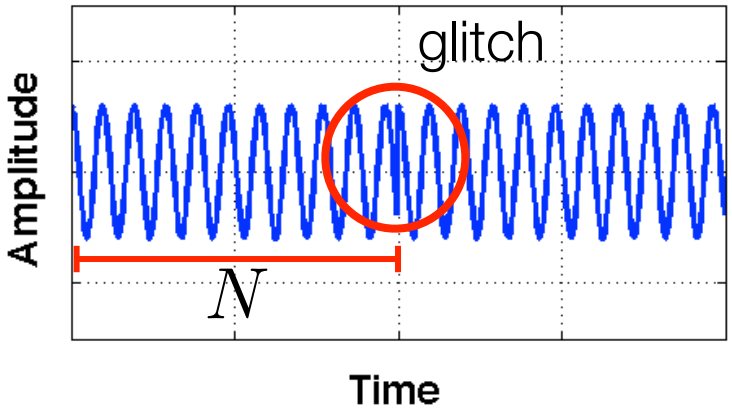
Spectral Leakage

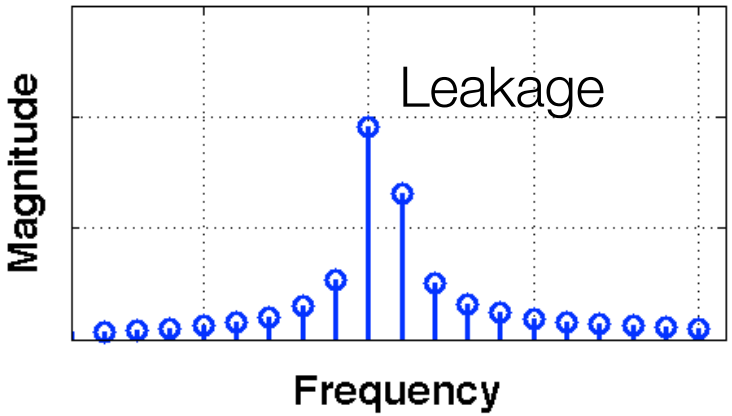
Whole number of periods



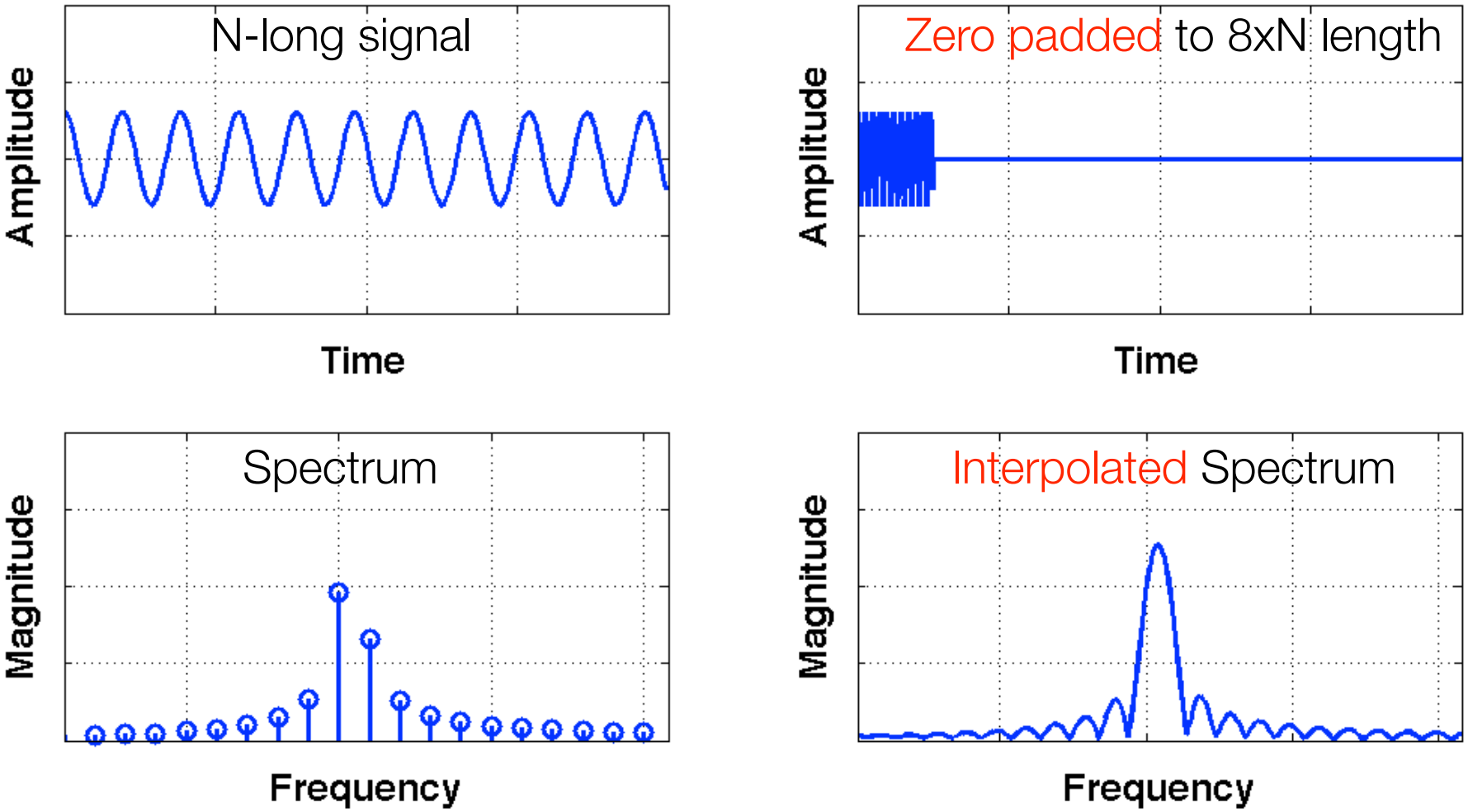


Fractional number of periods





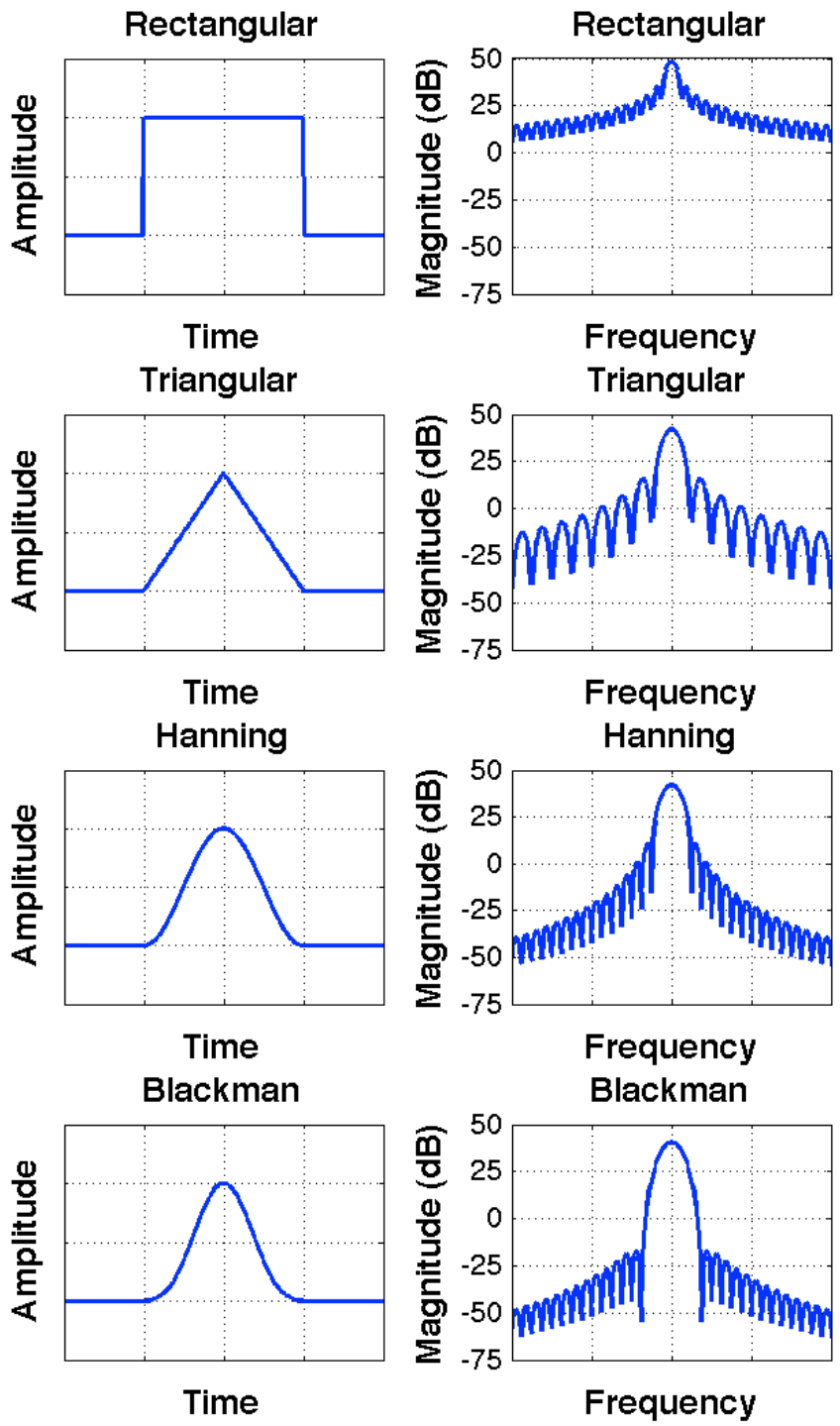
Zero padding



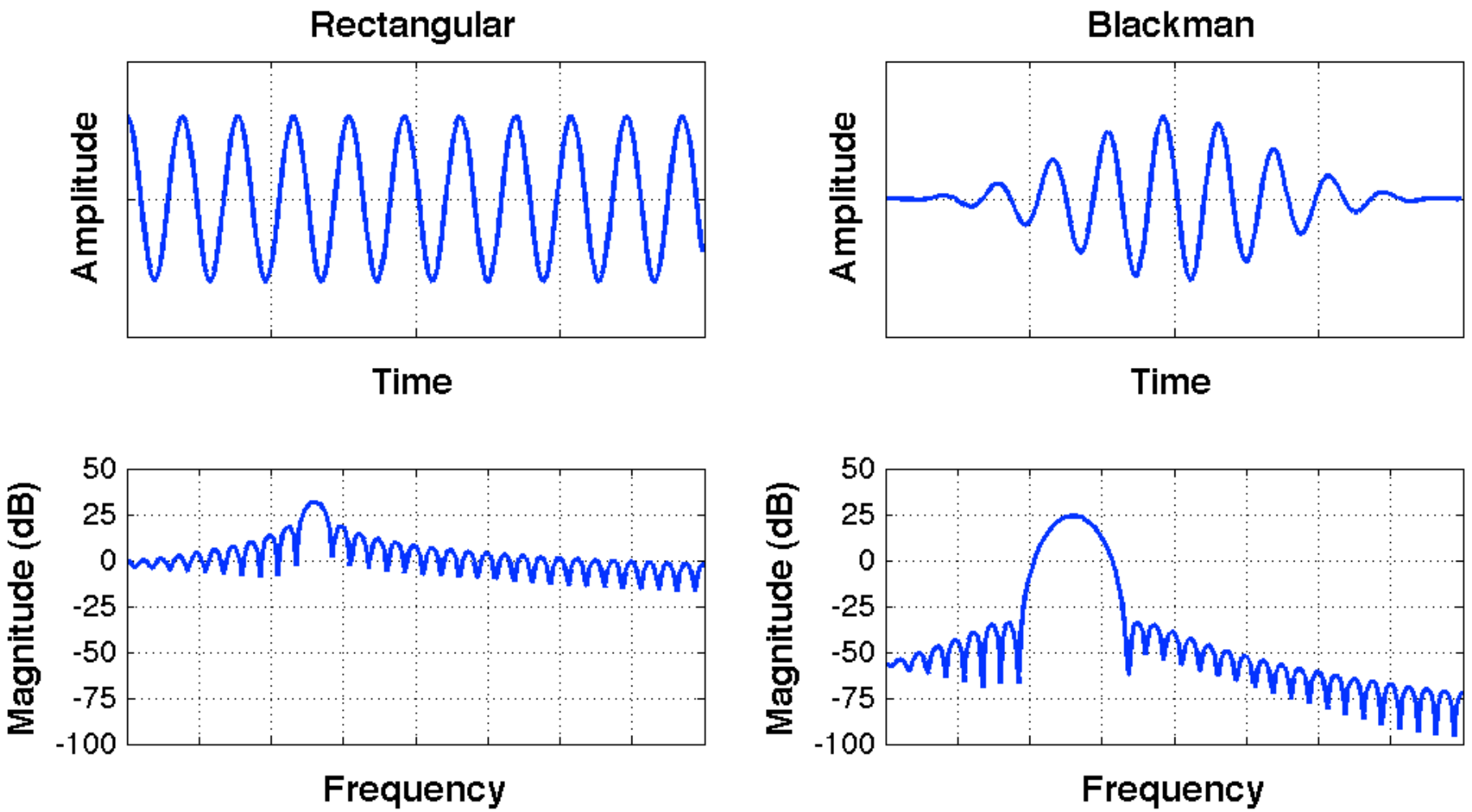
Windows

- We are effectively using a rectangular window: w(n)
- Spectrum = convolution of X(k) and W(k)
- Ideal window: narrow central lobe; strong attenuation in sidebands
- Figures show Magnitude in dB:

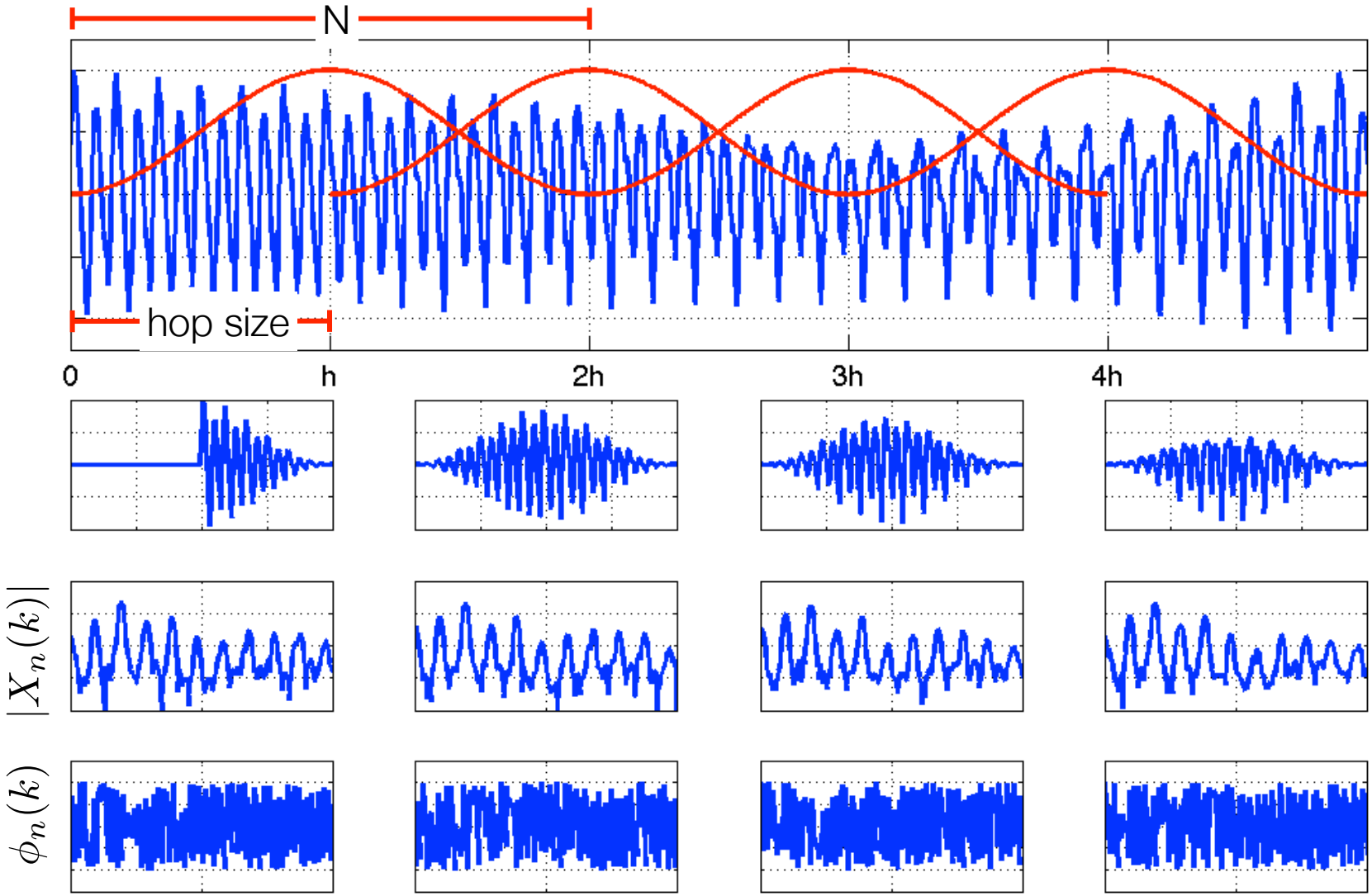
$$dB(X) = 20 \times log_{10}(X)$$



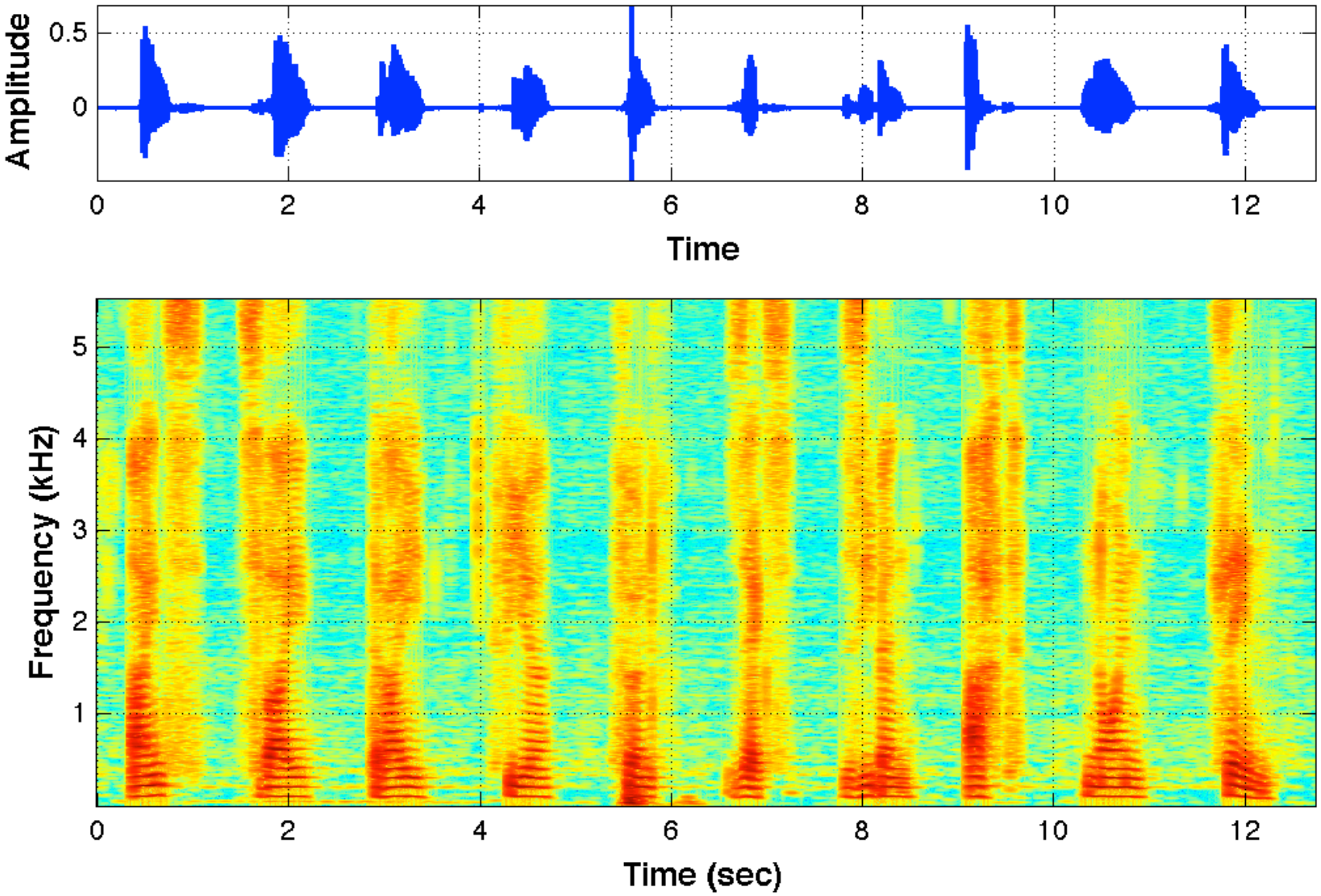
Windowing


Rectangular
Amplitude
Time
Blackman
Amplitude
Time
Magnitude (dB)
50
25
0
-25
-50
-75
-100
Frequency
Magnitude (dB)
50
25
0
-25
-50
-75
-100
Frequency

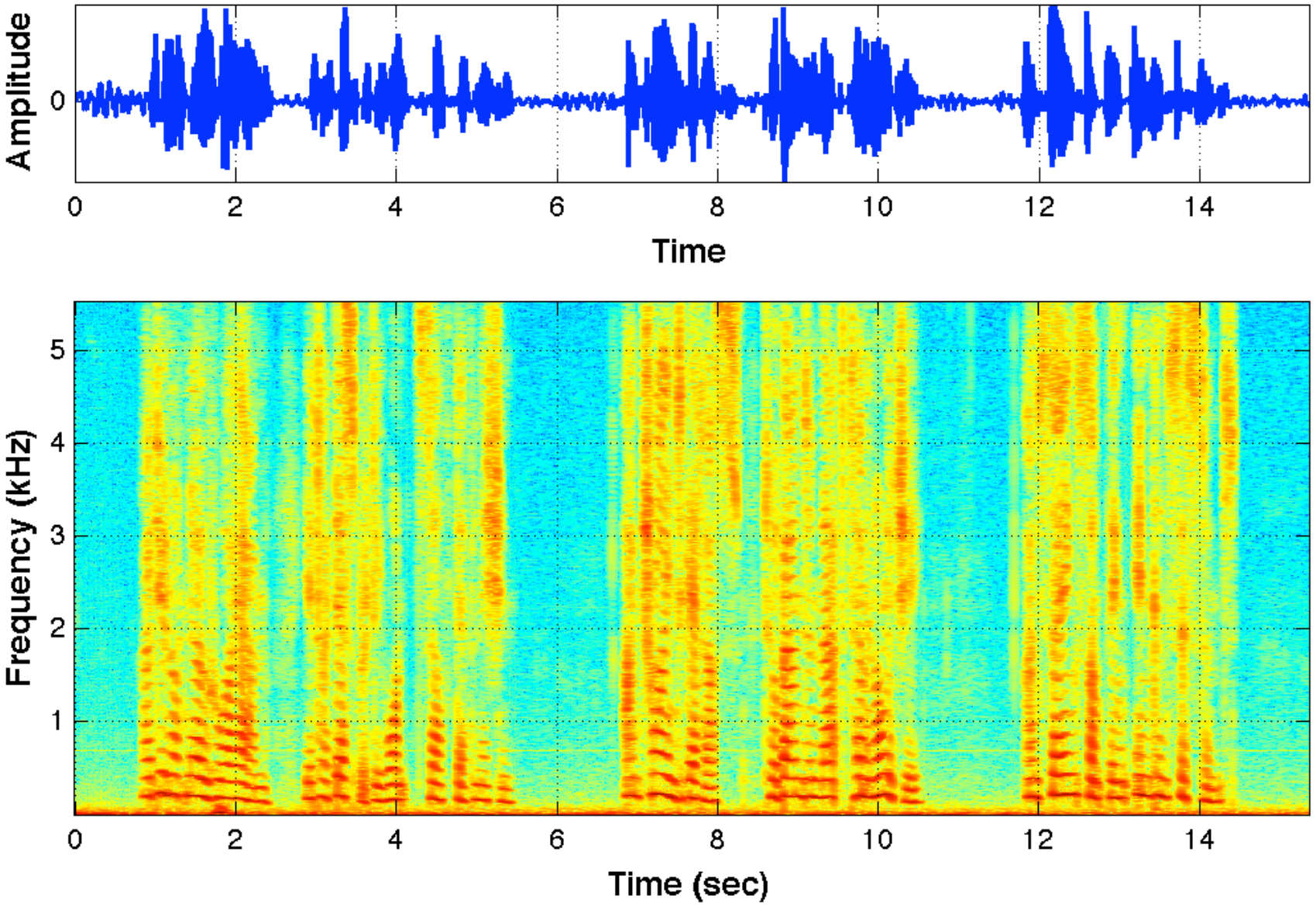
Short-Time Fourier Transform (STFT)



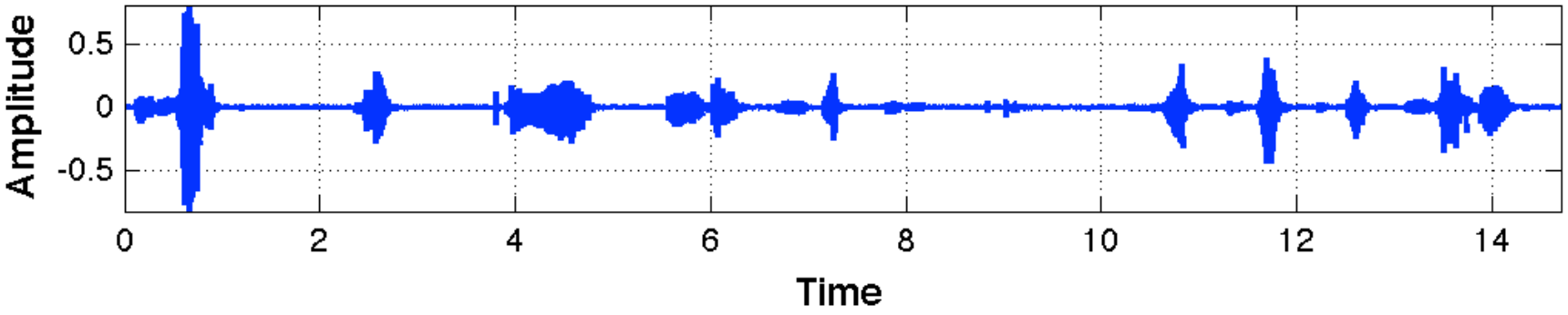
Spectrogram - male speaker

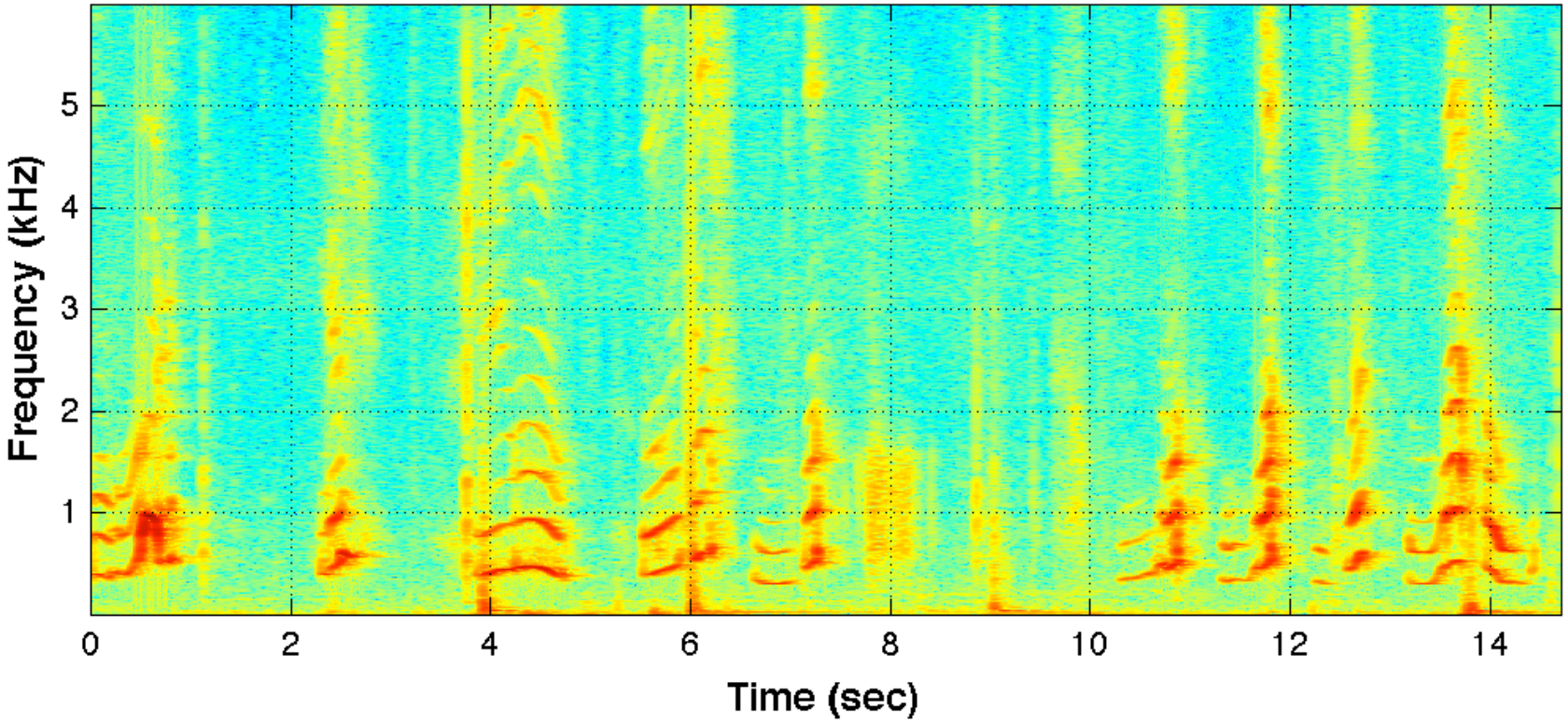


Spectrogram - female speaker

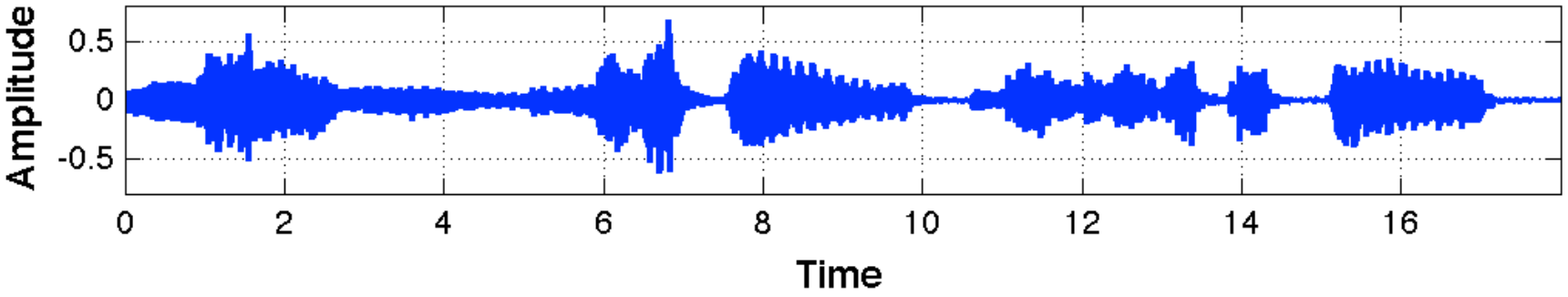


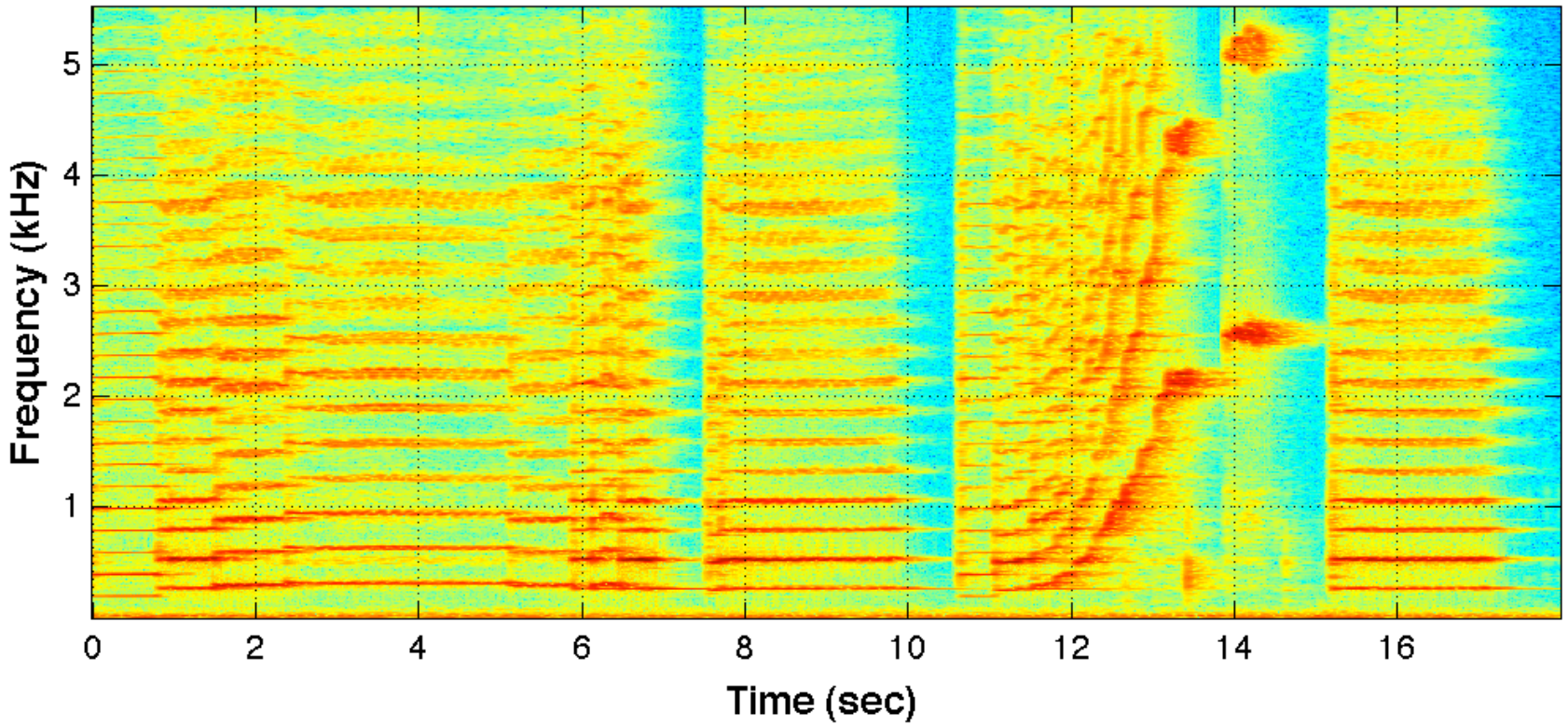
Spectrogram - baby cooing



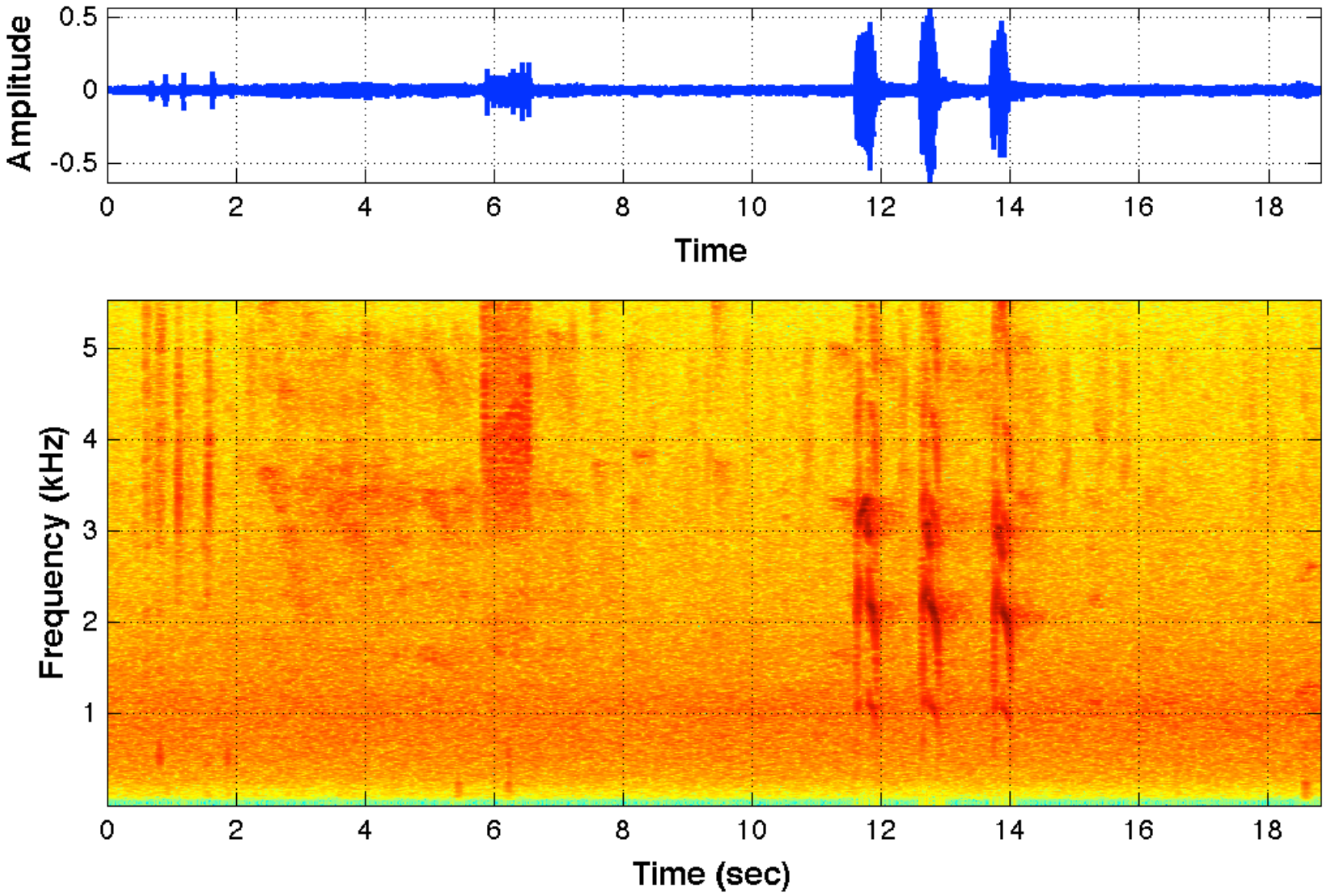


Spectrogram - violin

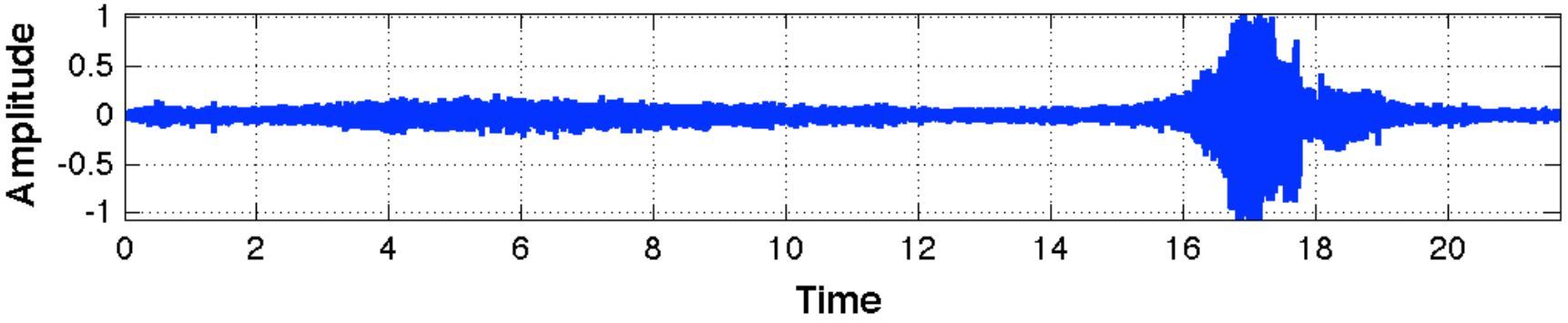


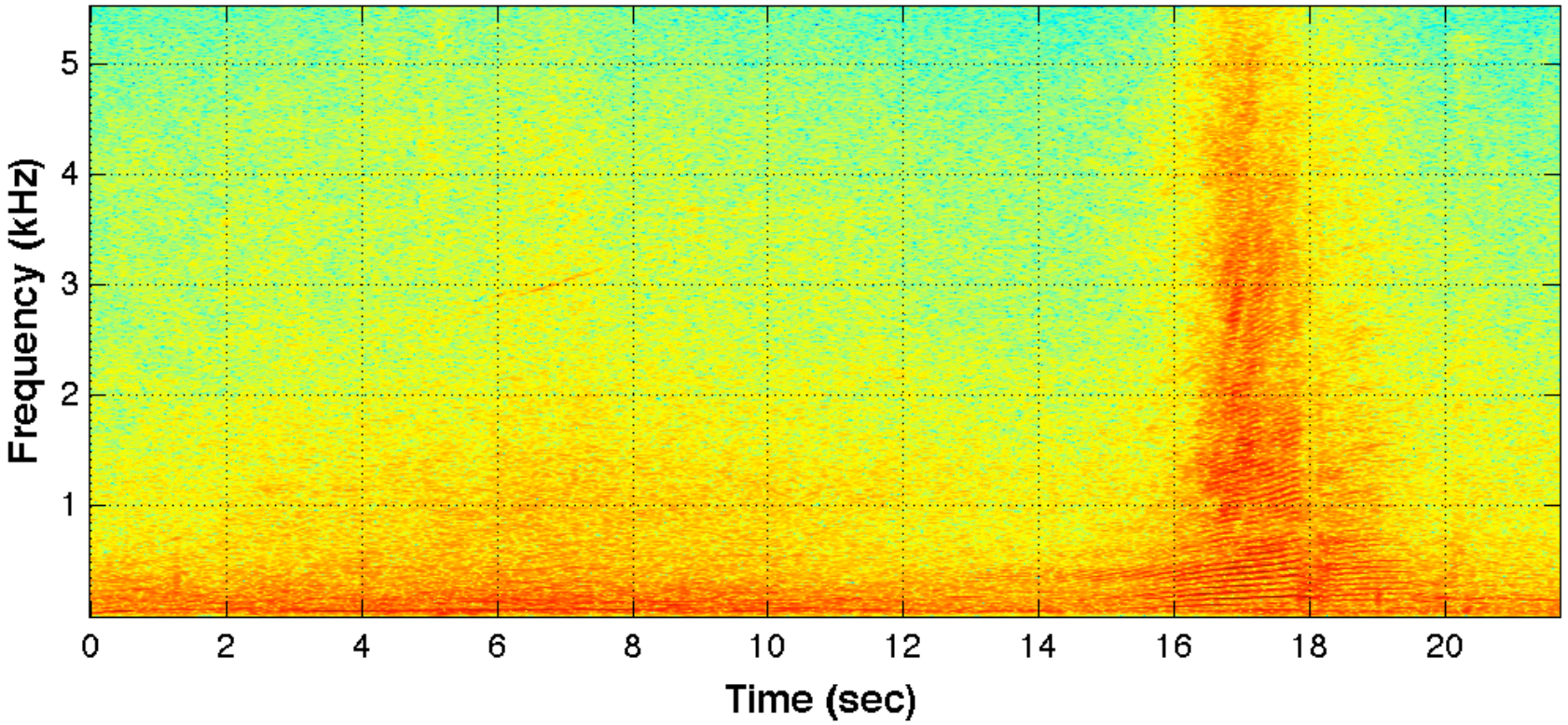


Spectrogram - birds by lakeside

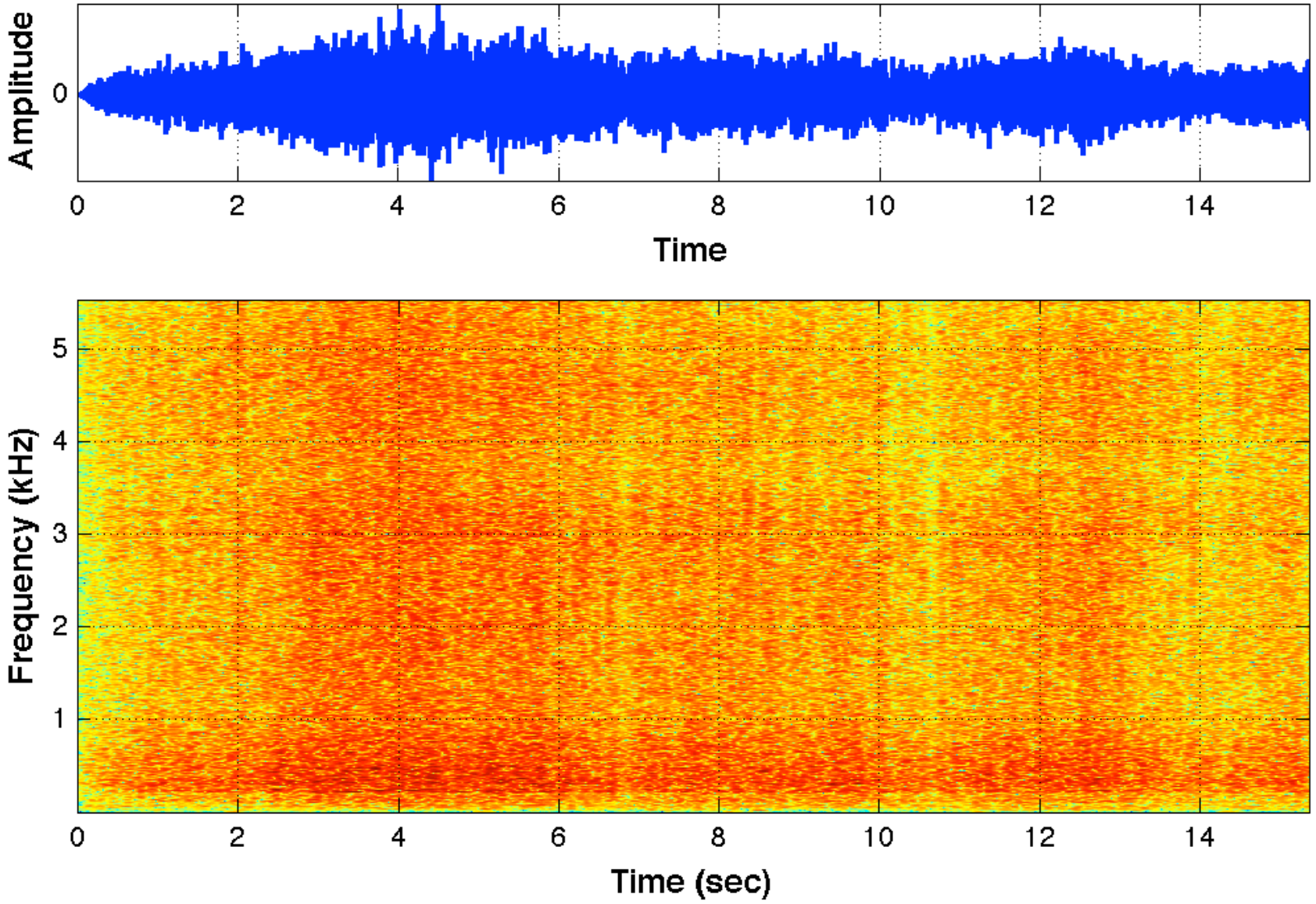


Spectrogram - street

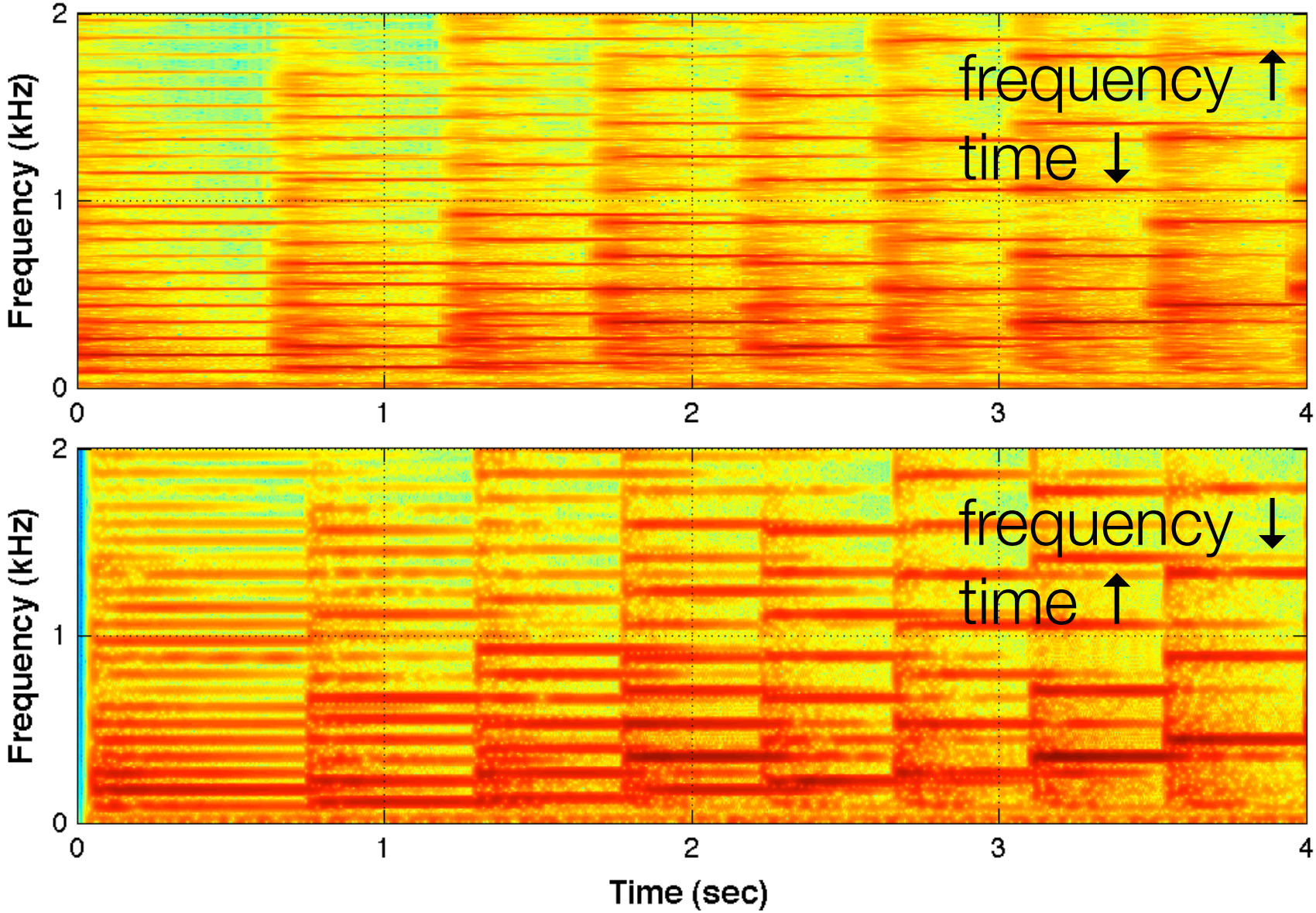




Spectrogram



Time vs Frequency Resolution



Instantaneous Frequency

The instantaneous frequency for frequency bin k at time instant mh can be defined as (Arfib et al., 2003):

$$f_i, k(m) = \frac{1}{2\pi}\frac{d\phi_k(m)}{dt} = \frac{1}{2\pi}\frac{\Delta\phi_k(m)}{h/f_s}$$

where,

$$\Delta\phi_k(m) = \Omega_k h + princarg[\phi_k(m) - \phi_k(m-1) - \Omega_k h]$$

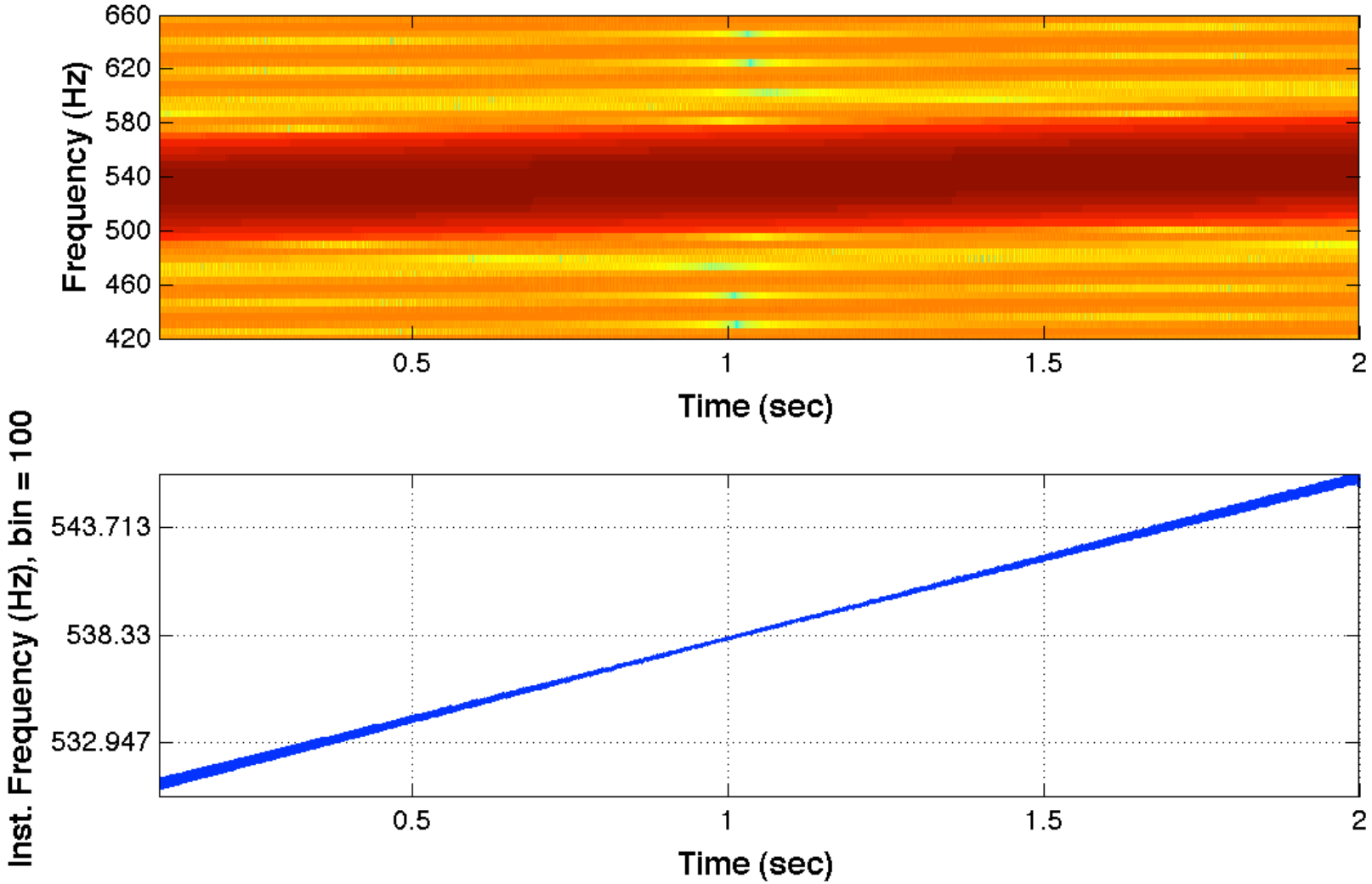
$$\Omega_k = \frac{2\pi k}{N}$$

and

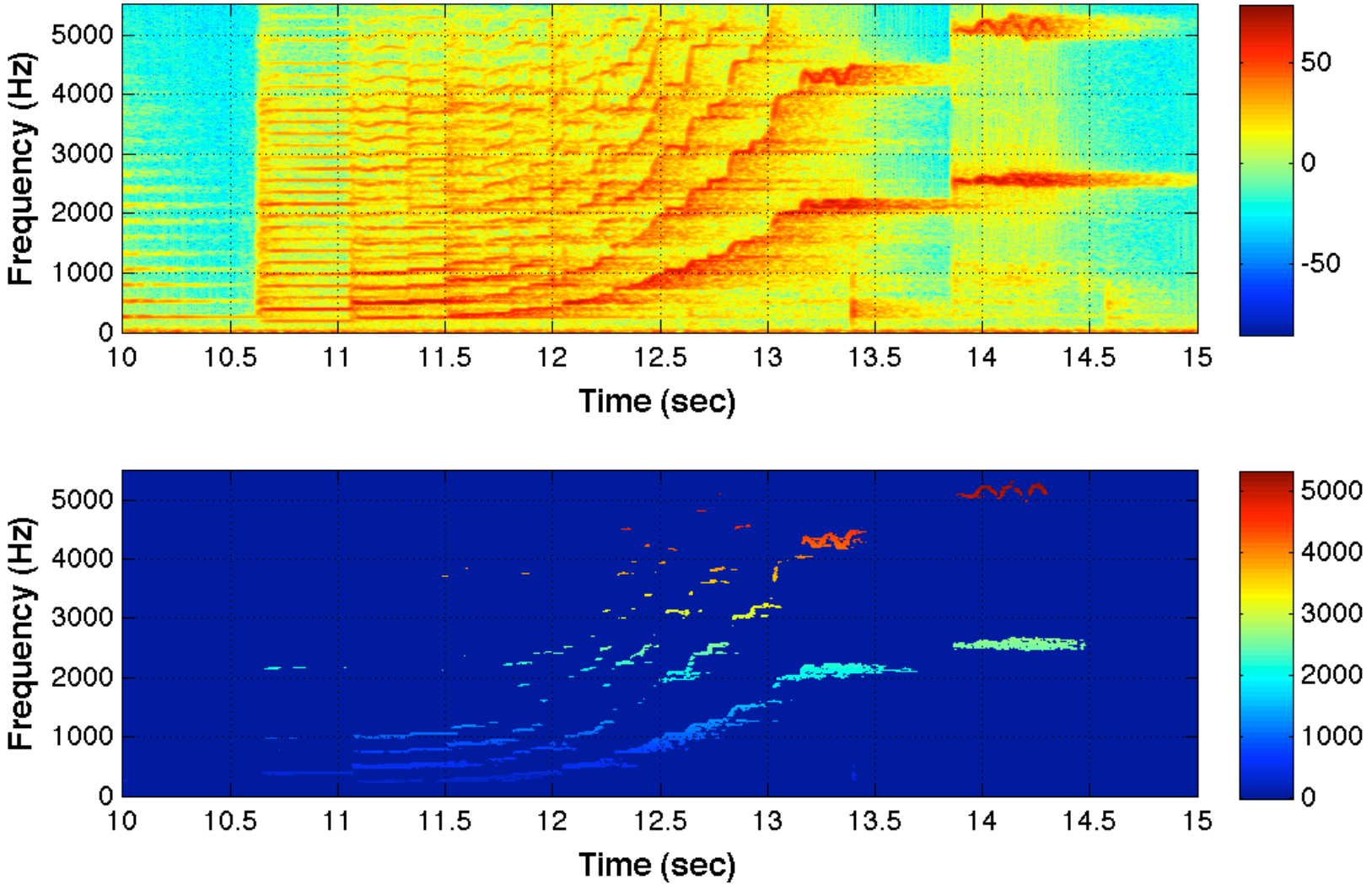
$$princarg(x) = \pi + [(x+\pi) mod (-2\pi)]$$

wraps the phase to the $(-\pi, \pi]$ range.

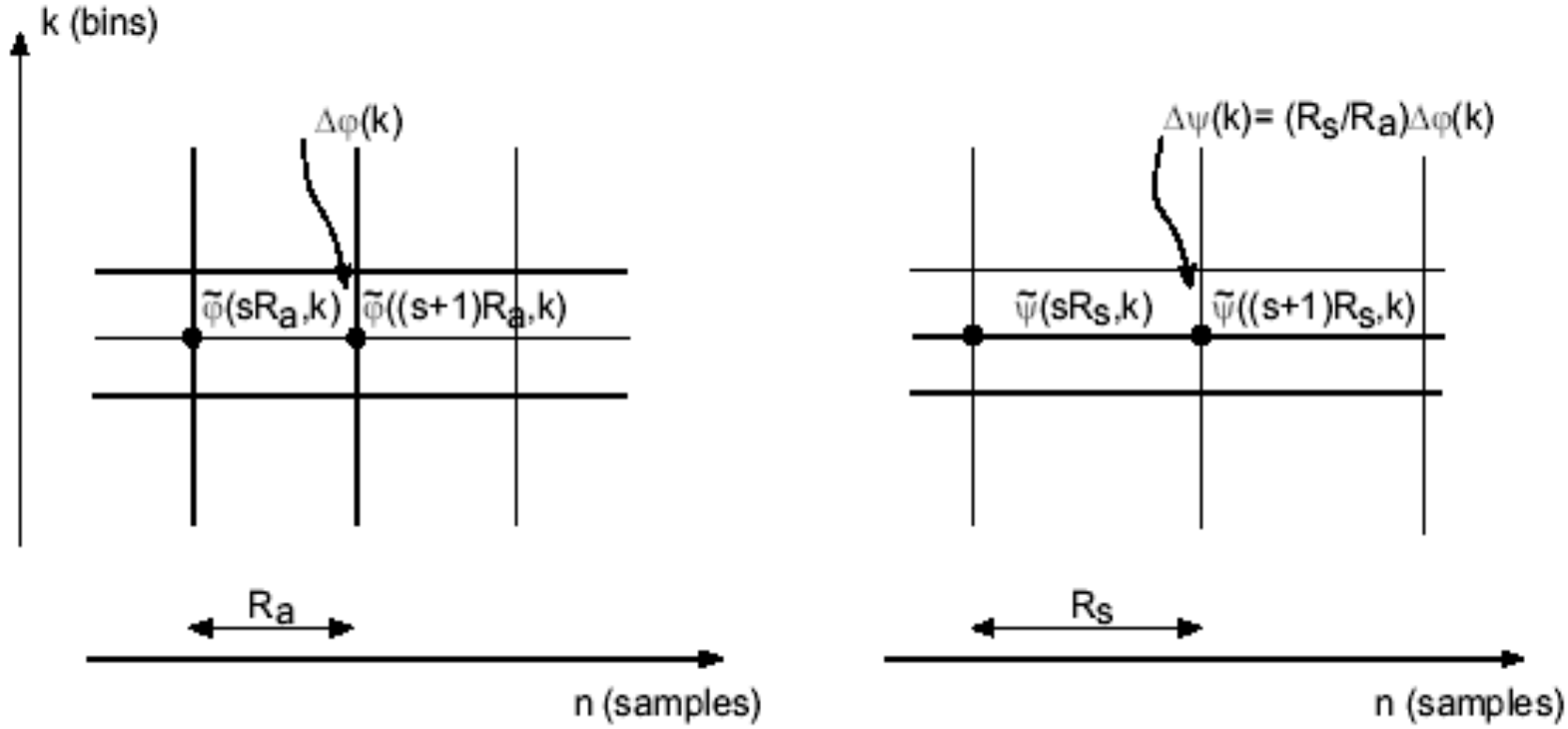
Instantaneous Frequency



Instantaneous Frequency



Time Scaling



* from DAFX book, chapter 8

- Solo guitar/polyphonic examples: (1) original, (2) standard, (3) adaptive

Sinusoidal Modeling

- The signal is approximated as a sum of time-varying sinusoidal components plus a residual:

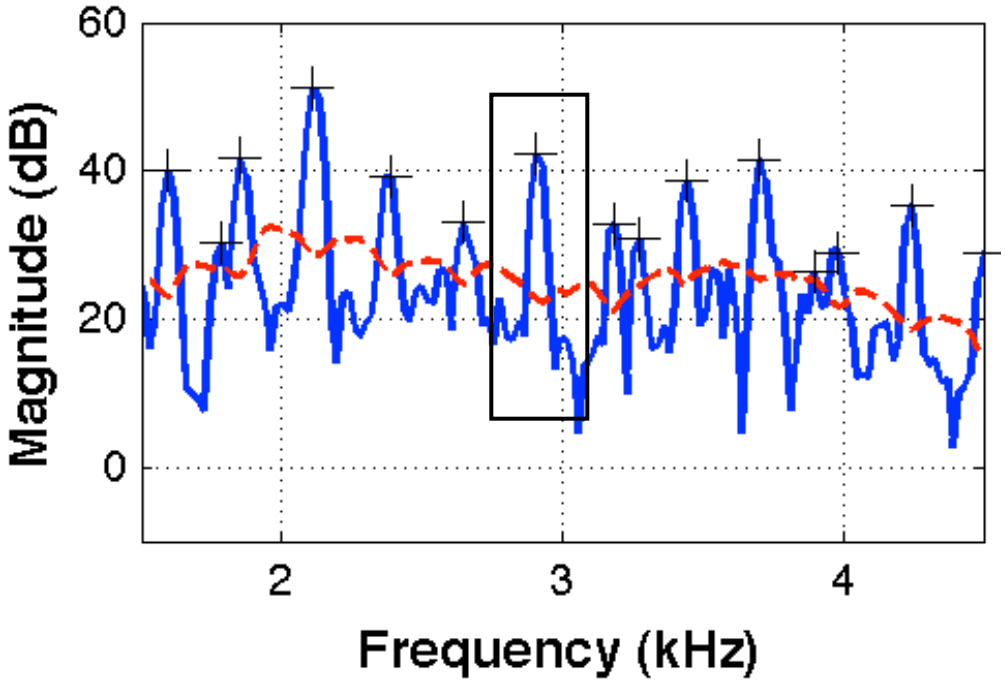
$$x(n) \approx \sum_{k=0}^{K} a_k(n) cos(\phi_k(n)) + e(n)$$

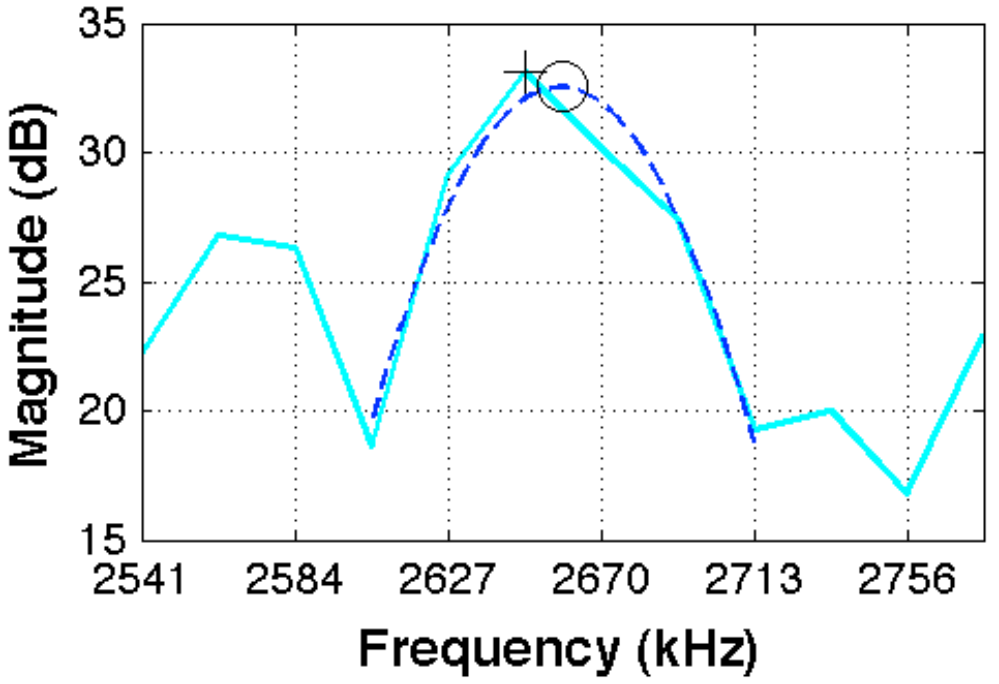
where,

$a_k = \text{instantaneous amplitude}$
$\phi_k = \text{instantaneous phase}$
$e(n) = \text{residual (noise)}$

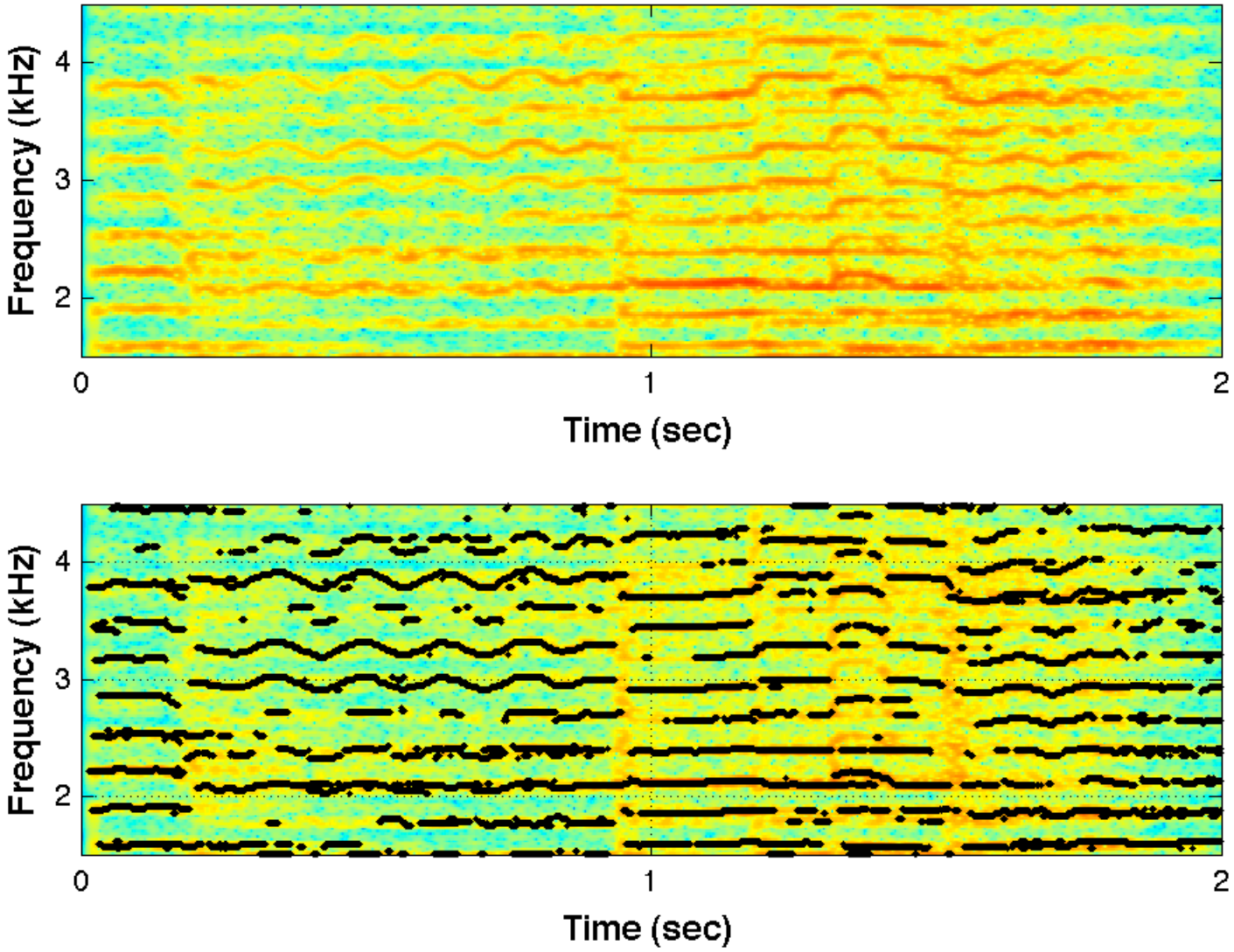
Peak picking + interpolation

- Sinusoidal components are peak-picked.
- Instantaneous magnitude and phase values are obtained by interpolation.
- Components are tracked over time

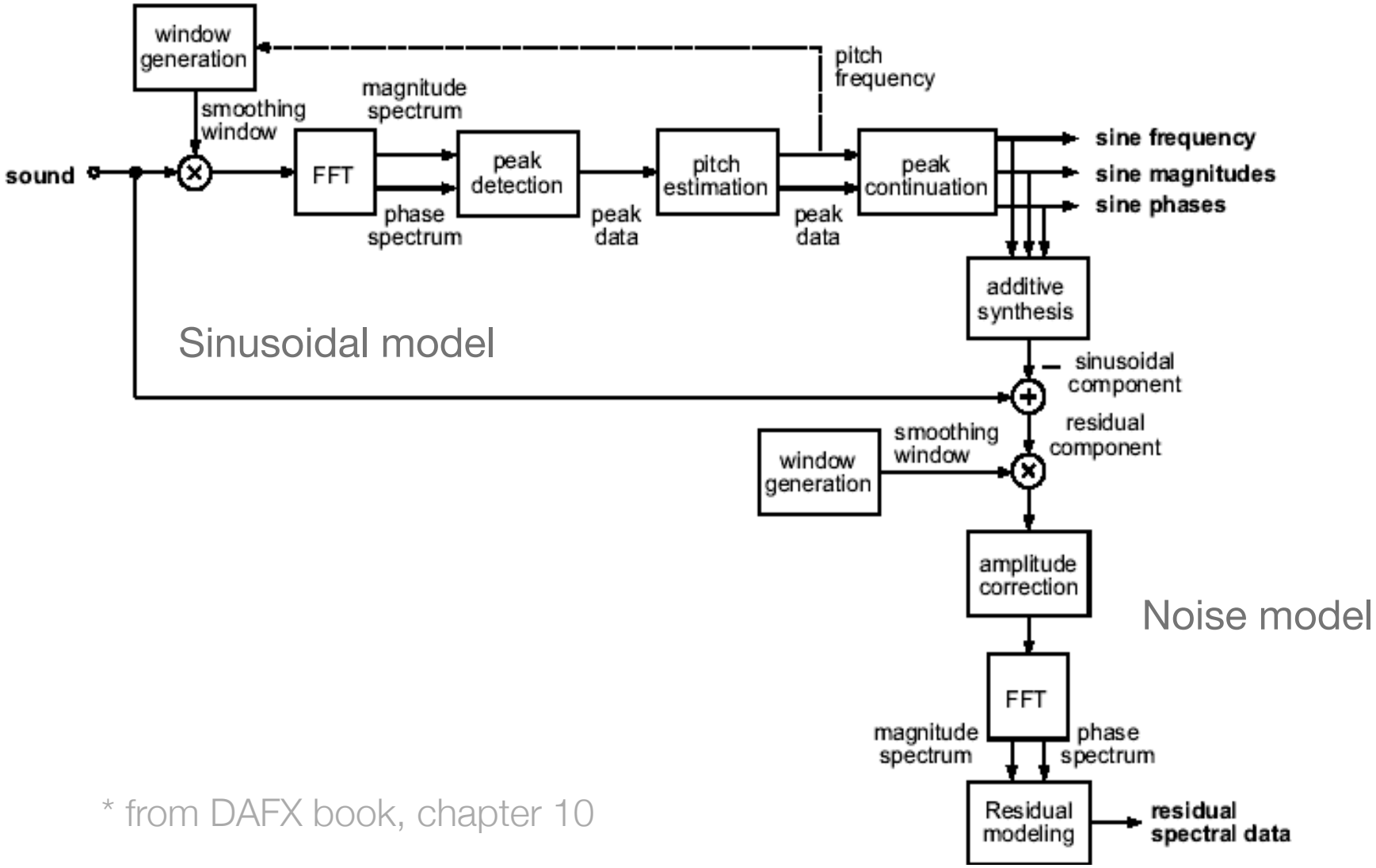




Sinusoidal Tracking



Sinusoidal Modeling

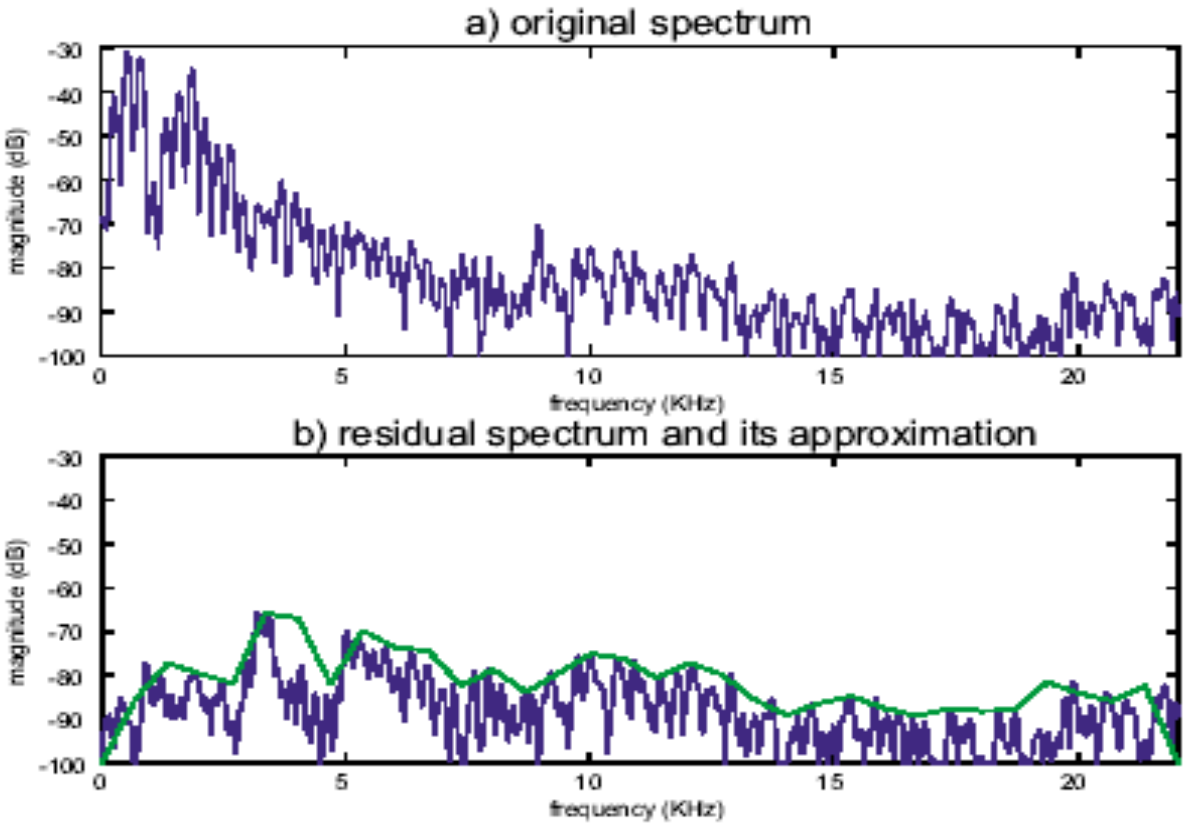


* from DAFX book, chapter 10

Noise Modeling

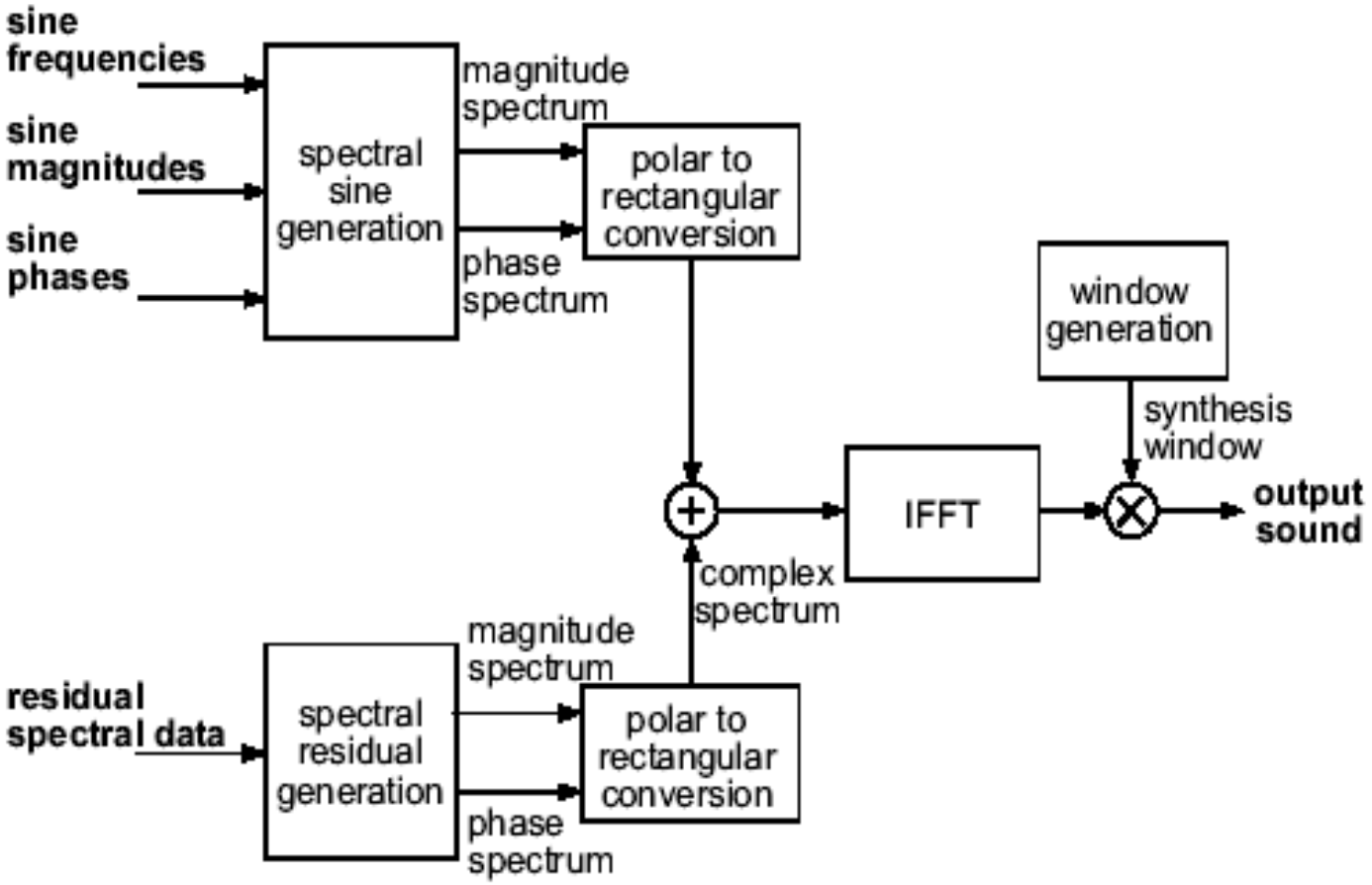
- We assume the residual to be a stochastic signal, i.e. can be described by its general spectral characteristics
- It is not necessary to maintain instantaneous phase or exact magnitudes.
- Hence, it can be modeled as the output of a time-varying filter driven by noise.
- The filter parameters encode the general spectral characteristics of the residual.
- Filter design usually involves an approximation of the spectral shape using: channel vocoder, LPC, Cepstrum, etc (see Lecture 5).

Noise Modeling



- Example approximation using max value per frequency band

Synthesis



* from DAFX book, chapter 10

Examples

http://mtg.upf.edu/technologies/sms?p=Sound%20examples

References

- Smith, J.O. “Mathematics of the Discrete Fourier Transform (DFT). 2nd Edition, W3K Publishing (2007)
- Zölzer, U. (Ed). “DAFX: Digital Audio Effects”. John Wiley and Sons (2002): chapter 8, Arfib, D., Keiler, F. and Zölzer, U., “Time-frequency Processing”; and chapter 10: Amatriain, X., Bonada, J., Loscos, A. and Serra, X. “Spectral Processing”.
- Pohlmann, K.C. “Principles of Digital Audio”. 6th Edition. McGraw Hill (2011): chapter 2.
- Loy, G. “Musimathics, Vol.2”. MIT Press (2011): chapter 3.
- Francis Rumsey and Tim McCormick (2002). “Sound and Recording: An Introduction”, Focal Press. (Chapter 1)
- Mitra, S. (2005). “Digital Signal Processing”. McGraw-Hill Science/Engineering/Math; 3rd edition